

THE ARITHMETIC OF CRITICAL VALUES II: CRITICAL ELLIPTIC CURVES

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ABSTRACT. In this second chapter of the *Arithmetic of critical values* series (ACV), we focus on certain elliptic curves E_f whose branch locus coincides with that of a quartic polynomial f . We give a direct proof of the fact, already shown non-constructively in ACV I, that these curves have a 3-isogeny. Our methods are Galois-theoretic, and lead us to study some other curves naturally attached to f . In the second part of the paper, we focus on the interplay between these families of curves, and on some remarkable arithmetic properties they exhibit. Following our constructions, we give applications in the theory of decomposable Jacobians, of Selmer and Tate-Shafarevich groups of elliptic curves, as well as some dynamical and Diophantine ones.

1. INTRODUCTION

In this work, we continue the study of the arithmetic of critical values of polynomials started in ACV I [35], focusing on the link between quartic polynomials and certain families of curves. For the moment we work with polynomials (and more generally morphisms of curves) defined over a field K of characteristic zero with algebraic closure k , which later on we will just take as a number field and $\overline{\mathbb{Q}}$, respectively.

In our previous work, we defined the moduli space $H_d \rightarrow \text{Conf}_d(\mathbb{P}_k^1)$ of degree d polynomials with given critical values, mainly by appealing to the theory of Hurwitz spaces. In general, the latter classify (G) -covers $Y \rightarrow X$ of a fixed curve X (up to (G) -cover isomorphism) having given branch locus and local monodromy data. So, Hurwitz spaces relative to X are naturally covers of the configuration space $\text{Conf}_d(X)$.

We also defined the *reduced* Hurwitz space $\mathcal{H}_d \rightarrow \text{Conf}_d(\mathbb{P}_k^1)/\text{PGL}_2(k)$ of degree d polynomials. Reduced Hurwitz spaces classify covers as above up to the natural action of $\text{Aut}(X)$ —which naturally carries over to the branch locus. As it turns out, *Morse* polynomials (see [43, §4.4] or §2 for the definition) of degree d are the only covers $f : \mathbb{P}_k^1 \rightarrow \mathbb{P}_k^1$ simply ramified at $d - 1$ points, fully at one point and with monodromy group S_d , once we require the d -cycle to be at $f^{-1}(\infty) = \{\infty\}$. Therefore, after the $\text{PGL}_2(k)$ -quotient, we get (see [35, §3.2]) a moduli space whose k -points are in bijection with linear equivalence classes of polynomials having given $\text{PGL}_2(k)$ -class of critical values, where we define:

Definition 1.1. *Two polynomials $f, g \in k[t]$ are **linearly equivalent** if there exist affine transformations $\alpha, \gamma \in \text{Aff}(k)$ such that $f \circ \alpha = \gamma \circ g$.*

$\mathcal{H}_d$ turns out to be an affine algebraic variety of dimension $d - 3$ (see [10, §1-2] for more details) and, under the aforementioned bijection, any $P \in \mathcal{H}_d(K)$ has a polynomial representative defined over an extension of K of degree bounded by a function of d ; this extension can be taken as K itself

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when $d = 4$ [35, Theorem 4]. Coming from a cover of $\text{Conf}_4(\mathbb{P}_k^1)$, $\mathcal{H}_4$ is a modular curve. In ACV I [35, Lemma 4.1], we showed that

$$(1) \quad \mathcal{H}_4 \simeq_{\mathbb{Q}} Y_0(3).$$

This has among its consequences the existence of infinitely many pairs of linearly inequivalent polynomials over $\mathbb{Q}$ with the same critical values [35, Theorem 1], which was the main application for the first chapter of the ACV series.

1.1. Structure of the paper and main results. In §2 we recall some basic facts about critical values, along with more details on the correspondence between linear classes of quartic polynomials and elliptic curves with a 3-isogeny. We then answer some natural questions left open after ACV I. Our previous proof of (1) relied on the Cummins-Pauli classification for genus-0 subgroups of $\Gamma(1)$, but one would naturally seek a direct proof of this isomorphism. In §3 we obtain such a proof by studying the geometry of the Galois closure of $f : \mathbb{P}_k^1 \rightarrow \mathbb{P}_k^1$, which is a curve of genus 4. In particular, we prove (cf. Proposition 3.3 and Theorem 3.5):

Theorem 1. *Let K be a number field and let $f \in K[X]$ be a quartic polynomial whose critical points are distinct and do not form an arithmetic progression. The Galois closure of the cover $f : \mathbb{P}^1 \rightarrow \mathbb{P}^1$ factors through a 3-isogeny of elliptic curves over K .*

In §4 we investigate the arithmetic and geometry of various families of curves that arise in the Galois diagram of Theorem 1, including that of *critical elliptic curves* E_f , a genus-2 family whose Jacobian splits as $E_f \times E'_f$, and the complementary (also called Prym) family E'_f . We show that these families are fibrations over a suitably punctured projective line, and that the Prym family has a nontorsion section. The latter will be central in the many of the Diophantine applications of the later sections.

§5 is devoted to the local arithmetic of critical elliptic curves, which together with the other families exhibit many remarkable properties. In particular, we construct the first infinite family of non-isogenous 2-Selmer companions [33] (cf. Theorem 5.5):

Theorem 2. *There exist infinitely many pairs (E_1, E_2) of non-isogenous rational elliptic curves that are 2-Selmer companions over $\mathbb{Q}$, and such that no two pairs are related by quadratic twisting.*

We also exhibit many curves with 4-torsion in their Tate-Shafarevich group isomorphic to $(\mathbb{Z}/4\mathbb{Z})^2$, and prove visibility [14] for the resulting order-4 elements.

In §6 we turn to global Diophantine applications. We settle, over a fixed quadratic extension, the infinitude question for rational solutions to the same-polynomial variables separated equation (cf. Theorem 6.2):

Theorem 3. *Let f be a quartic with rational coefficients. The equation*

$$(2) \quad f(X) = f(Y), \quad X \neq Y$$

has infinitely many solutions over $\mathbb{Q}(i)$, unless the j -invariant of the critical points of f is $0, -27648/11$ or $55296/5$.

We also construct three rational genus-4 curves whose Jacobian splits over $\mathbb{Q}$ as the fourth power of an elliptic curve; two are novel, while the other recovers Bring's curve, the only previously known such example:

Theorem 4. *For f equal to the polynomials*

$$X^4 + 30X^2 + 64X, \quad X^4 + 90X^2 + 72X, \quad X^4 + 10X^2 + 40X,$$

the Jacobian of the Galois closure of $f : \mathbb{P}^1 \rightarrow \mathbb{P}^1$ is isogenous over $\mathbb{Q}$ to the fourth power of

$$y^2 = x^3 - 75x - 506, \quad y^2 = x^3 + 45x - 18, \quad y^2 = x^3 - 675x - 79650,$$

respectively.

In the same section, we give more examples of interesting properties of the curves attached to critical values, for instance in terms of ranks and of Tate-Shafarevich groups.

2. CRITICAL VALUES OF QUARTIC POLYNOMIALS

In order to make this work self-contained, we briefly recall some basic facts about polynomials and their critical values. Still, this work is intended as the sequel to [35], to which we kindly point the reader interested in more detail.

2.1. Basics on critical values. Recall that, for a polynomial $f \in k[x]$ of degree d , the critical values C_f are the branch points of the induced map $f : \mathbb{P}_k^1 \rightarrow \mathbb{P}_k^1$. They consist of ∞ —over which there is total ramification—and the points in $f(Z_f)$, where the Z_f is the (multi)set of roots of f' , i.e. the finite ramification points. If $x \in Z_f$, we call x a *critical point* of f . Let us remark that the set of monic polynomials $f \in k[t]$ of degree d and such that $f(0) = 0$ can be identified with $\mathbb{A}_k^{d-1}$, via the bijection $f \mapsto Z_f$. The group of k -linear maps $\text{Aff}(k) = \{z \mapsto az + b, (a, b) \in k^\times \times k\}$ acts coordinate-wise on $\mathbb{A}_k^{d-1}$, and we have

$$(3) \quad Z_{\alpha \circ f} = Z_f, \quad C_{\alpha \circ f} = \alpha(C_f)$$

for any $\alpha \in \text{Aff}(k)$; so, restricting to monic polynomials with zero constant term loses no information about critical points and values. In the (generic, under the above identification) locus where critical points are distinct, a simple application of the Riemann-Hurwitz formula yields

$$e_x(f) = 2$$

for any $x \in Z_f$. Moreover, as shown in [35, Lemma 2.2], critical values are also generically distinct; following [43, p. 4.4], we refer to a polynomial with distinct critical values as *Morse*.

In addition to (3) we also have

$$(4) \quad Z_{f \circ \alpha} = \alpha^{-1}(Z_f) \text{ and } C_{f \circ \alpha} = C_f$$

for any $\alpha \in \text{Aff}(k)$. The map sending the $\text{Aff}(k)$ -class of $\hat{x} \in \mathbb{A}_k^{d-1}$ to that of $\hat{y} \in \mathbb{A}_k^{d-1}$ such that there is $f \in k[x]$ with $Z_f = \hat{x}$ and $C_f = \hat{y}$ is, therefore, well-defined. It is a generically étale morphism of algebraic varieties defined over $\mathbb{Q}$ [35, §2.1]. When $d = 4$, it is a map between two curves, since the quotient by the affine relation reduces the dimension by 2. The base curve is naturally $Y(1)$, with coordinate the j -invariant j_{CV} of the critical values. In [35] we showed that the covering curve—the (reduced) Hurwitz space of quartic polynomials—is then simply the modular curve $Y_0(3)$ (or $Y_1(3)$ —they are the same curve over $\mathbb{Q}$, see [35, Corollary 4.2.2]) with coordinate the j -invariant j of the quadruple of critical points (which here are taken to include ∞ , see [35, §3.1]). The above morphism is then given by

$$(5) \quad j_{\text{CV}} = 2^{-18} \frac{j(j - 1536)^3}{j - 1728} = \pi_3 \left(\frac{j}{64} - 27 \right),$$

where $\pi_3 : X_0(3) \rightarrow X(1)$ is the classical Hauptmodul map $u \mapsto \frac{(u+3)^3(u+27)}{u}$ [49, p.23]. This is shown in [35, Lemma 4.3].

2.2. The case of quartics.

Definition 2.1. We call a quartic $f \in k[X]$ **good** if $j(Z_f) \notin \{1728, \infty\}$, otherwise we call f **bad**.

Remark 2.1.1. (i) By (5), a quartic is good if and only if the j -invariant of its critical values lies in k (that is, it is not ∞).

(ii) The bad quartics with $j(Z_f) = 1728$ are those linearly equivalent (over k) to the Chebyshev quartic $8X^4 - 8X^2 + 1$, while those with $j(Z_f) = \infty$ are those linearly equivalent to $X^4 + X^3$ or X^4 . This follows, for instance, from the computations at the end of [35, §5.2].

Definition 2.2. Given a quartic $f \in K[X]$, the **critical curve relative to f** is the projective curve over K with affine model

$$(6) \quad E_f : y^2 = \text{Disc}_X(f(X) - x).$$

Lemma 2.3. The curve E_f is smooth if and only if f is good, in which case it is an elliptic curve and we have $E_f[2] = C_f$. With our choice of normalization for the quartics, a Weierstrass model for E_f over K is given by

$$(7) \quad -y^2 = \prod_{v \in C_f} (x - v).$$

Proof. If f has leading coefficient A , $\text{Disc}_X(f(X) - x)$ is the cubic polynomial in x with leading coefficient $-2^8 A^3$ which vanishes at those $x_0 \in k$ for which the polynomial $f(X) - x_0$ has a repeated root, i.e. the critical values of f . The claim then follows from Remark 2.1.1(i) after sending $y \mapsto 16y$. $\square$

Remark 2.3.1. (1) If f is bad, a direct computation with the representatives in Remark 2.1.1 (ii) shows that E_f is a nodal cubic, except for the case of the linear equivalence class of $f(t) = t^4$.

(2) If f is good we have $j(E_f) = j_{\text{CV}} = \pi_3 \left(\frac{j(Z_f)}{64} - 27 \right)$, in virtue of Lemma 2.3 and (5).

In the remainder of this work, we normalize our quartics $f \in K[X]$ to be monic and have no cubic nor constant terms: in formulas, $f(X) = X^4 + 2AX^2 + 4BX$. This can be achieved with a K -linear change of variable followed by a K -linear rescaling of the polynomial, so it has no impact on the equivalence classes of critical points and values, by (3) and (4). Thanks to the interpretation of $Y_0(3)$ as the moduli space of elliptic curves with a 3-isogeny, our correspondence can be seen in the following way:

Proposition 2.4. Let E/K be an elliptic curve with $j = j(E) \neq 1728$, and let $Y^2 = q(X)$ be a short Weierstrass equation for E . Define the quartic $f \in K[X]$ by

$$(8) \quad f(X) = f_E(X) := \int_0^X q(u) du.$$

Then, the critical elliptic curve E_f has a 3-isogeny over K .

Remark 2.4.1. At this stage it might be worth mentioning that the polynomial f_E appearing in (8) differs from $\frac{\psi_3(E)}{12}$ by an additive constant, where $\psi_3(E)$ is the 3-division polynomial of E —normalized, as usual [45, III, Exercise 3.7], with leading coefficient 3. Moreover, $E_{-\psi_3(E)}$ is one of the (four, at most) quadratic twists of E_f with a 3-torsion point over K , and not just a 3-isogeny. One may wonder whether the covert appearance of the 3-division polynomial is related to our Hurwitz space being isomorphic to $Y_1(3)$; such an explanation does not appear to be immediate.

2.3. Some natural questions. Picking up from [35, Remark 2.1], we now pose and answer some questions about critical values in degree 4. For instance, it can be natural to ask which good quartics $f \in K[X]$ have one (resp. three) K -rational critical value(s). Looking at the G_K -action on the equations [35, (4)] defining them, one immediately sees that this is equivalent to f having one (resp. three) K -rational critical point(s), which it maps to the K -rational critical value(s). Therefore, such f are simply those with $j(Z_f)$ in the image of the modular cover $X_1(2)(K) \rightarrow X(1)(K)$ (resp. $X(2)(K) \rightarrow X(1)(K)$).

In virtue of the classification and applications of equicritical quartics obtained in [35], it may be of some interest to consider the same question while requiring our f with one (resp. three) critical value(s) in K to also be equicritical to a quartic $g \in K[X]$ not linearly equivalent to it (an “equicritical twin”). From [35, Lemma 4.1] we know that quartics with an equicritical twin are rationally parametrized (up to acting linearly on the variable and the polynomial) by

$$Y_3 := Y_0(3) \times_{Y(1)} Y_0(3) \setminus \Delta,$$

where Δ is the diagonal in the fiber product. Therefore, all such f can be generated from a set of representative quartics with $j(Z_f) \in Y(K)$ by acting linearly on such quartics and the variable, where $Y := Y_3 \times Y_1(2)$ (resp. $Y := Y_3 \times Y(2)$) and the fiber product is over $Y(1)$. In the first case, the genus of the resulting cover of the j_{CV} -line $Y(1)$ is 0 and hence, as $x^4 - x$ is such a quartic (see [35, Theorem 1]), there are infinitely many such f over any number field K . More interesting is the case in which we require all critical values to be in K :

Proposition 2.5. *Let K be a number field. The following are equivalent:*

- *there exist infinitely many linearly inequivalent quartics $f \in K[X]$ with three K -rational critical values and which admit an equicritical twin in $K[X]$;*
- *the elliptic curve $E_0 : y^2 = x^3 + 1$ has positive K -rank.*

Proof. From the observations in the previous paragraph we find that a maximal set of such good f bijects with that of K -rational points of $Y_3 \times Y(2) \simeq Y_6$, where we again adopt Rubin and Silverberg’s notation from [40]. In [40, Theorem 2.1] they show that X_6 has affine model $y^2 = x^3 + 1$; since bad quartics account for finitely many cases up to equivalence, we are done by the Mordell-Weil Theorem. $\square$

Remark 2.5.1. We have $E_0(\mathbb{Q}) \simeq \mathbb{Z}/6\mathbb{Z}$, so there are finitely such f with rational coefficients. Using the parametrization [35, Theorem 1], any such good f would have to either share the same critical values of $x^4 - x$ —which are not all rational—or have, up to a linear change of variable, derivative $x^3 - 3tx - 2$ for some $t \in \mathbb{Q} \setminus \{-2, 0, 1\}$. It is not hard to see that the only such cubic to have three rational roots is $x^3 - 3x - 2 = (x - 1)^2(x + 2)$, so there are no such good $f \in \mathbb{Q}[x]$, and all six rational points of E_0 correspond to bad quartics or cusps introduced by compactifying the affine model of Y_6 .

Another question we can ask is which quartics $f \in \overline{\mathbb{Q}}[X]$ satisfy

$$(9) \quad Z_f = C_f.$$

This has, for instance, the application of exhibiting *post-critically finite (PCF)* [37] quartic polynomials, whose complete classification is still an open problem [.] We restrict to good quartics as the nontrivial case. Surely, equality of critical points and values implies equality of their j -invariants, so from [35, (21)] we obtain the equation

$$j = \frac{j(j - 1536)^3}{2^{18}(j - 1728)}$$

for the j -invariant of the critical points. This means that either $j \in \{0, \infty\}$ or $u = j - 1536$ satisfies $u^3 = 2^{18}(u - 192)$; observing that $u = 2^8 = 256$ is a solution, we find that such an f must have

$$(10) \quad j(Z_f) \in \{0, 1792, 128(11 \pm \sqrt{13})\}.$$

Now we have to look for f satisfying (9) inside each of the five linear equivalence classes satisfying (10). Take any such f with $j(Z_f) = j = j(C_f)$; then there exists $\alpha \in \mathrm{PGL}_2(\mathbb{C})$ mapping $Z_f \cup \{\infty\}$ to $C_f \cup \{\infty\}$ as sets. Up to post-composing α with the element $\beta \in \mathrm{PGL}_2(\mathbb{C})$ that fixes the set $C_f \cup \{\infty\}$ and swaps ∞ with $\alpha(\infty)$, we can assume that α is affine. Letting $f_1 = \alpha^{-1} \circ f$ and $f_2 = f \circ \alpha^{-1}$, we then have $Z_{f_i} = C_{f_i}$ in virtue of (3) and (4). Moreover, all quartics linearly equivalent to f satisfying (9) can clearly be obtained from this procedure for such an α , so their set is

$$\bigcup_{\alpha \in \mathrm{Aff}(\mathbb{C})} \alpha \mathrm{Stab}_{\mathrm{Aff}(\mathbb{C})}(Z_f) \circ f_1 \circ \mathrm{Stab}_{\mathrm{Aff}(\mathbb{C})}(Z_f) \alpha^{-1}.$$

Now, for $j \neq 0, 1728$ those stabilizers are trivial, as a quadruple with such a j -invariant is stabilized exactly by a Klein-four subgroup $C_2 \times C_2 \leq \mathrm{PGL}_2(\mathbb{C})$ acting faithfully on it. Therefore, for $j(Z_f) \in \{1792, 128(11 \pm \sqrt{13})\}$ there is exactly one such quartic up to the conjugation action of $\mathrm{Aff}(\mathbb{C})$. We can find a representative with minimal coefficient field as follows: take any elliptic curve $E_j/\mathbb{Q}(j)$ with j -invariant j and let X_i be its Weierstrass roots and f a primitive of its Weierstrass polynomial. Compute $x_i = f(X_i)$ and take an affine transformation $\alpha \in \mathrm{Aff}(\mathbb{Q}(j))$ mapping $\{X_i\}$ to $\{x_i\}$ (this exists by construction). Then, $\alpha^{-1} \circ f \in \mathbb{Q}(j)[X]$ is such a representative. Taking $j = 1792$ and $f(X) = X^4 - 42X^2 - 28X - 154$ —whose critical points are the Weierstrass roots of the elliptic curve with Cremona label 1764j1, having j -invariant 1792—we find that $C_f = 63Z_f - 448$. This yields a new example of quartic PCF conjugacy class:

Corollary 2.5.2. *The quartic $\frac{1}{63}(X^4 - 42X^2 - 28X + 294)$ is post-critically finite.*

For $j = 0$, every quartic $f \in K[X]$ is linearly equivalent over K to $f_B(x) = X^4 + 4BX$ for some $B \in K$, and equivalence classes correspond to those of B in $K^\times/(K^\times)^3$. We have $Z_{f_B} = \{\zeta, \zeta\omega, \zeta\omega^2\}$ where $\zeta^3 = -B$ and ω is a primitive third root of unity. On the other hand, $C_{f_B} = \{3B\zeta\omega^i, i = 0, 1, 2\}$. Therefore, (10) is satisfied by the polynomials $\alpha^{-1} \circ f_B$ and $f_B \circ \alpha^{-1}$ for $\alpha(t) = 3B\omega^i t$ and $i = 0, 1, 2$: the cases where $i = 1, 2$ correspond to the extra nontrivial affine automorphism of a quadruple with j -invariant 0. Focusing on rational quartics, we have showed:

Proposition 2.6. *Up to the conjugacy action of $\mathrm{Aff}(\mathbb{Q})$, the set of quartics $f \in \mathbb{Q}[X]$ with distinct critical points such that $Z_f = C_f$ has the following representatives:*

- $\frac{1}{63}(X^4 - 42X^2 - 28X + 294)$,
- $\frac{1}{3B}(X^4 + 4BX)$ for $B \in \mathbb{Q}^\times/(\mathbb{Q}^\times)^3$.

Remark 2.6.1. While for f_B of the second type we clearly have $f_B(X_i) = X_i$, so the critical points are preserved point-wise, it is easy to see that for $f(X) = \frac{1}{63}(X^4 - 42X^2 - 28X + 294)$ we have $f(X_i) = X_{\sigma(i)}$ for σ a 3-cycle.

Finally, in the spirit of [35, Theorem 2], we can ask *how likely* it is for a Galois orbit $\{x_1, x_2, x_3\}$ defined over K to be *critical over K* , that is, to consist of the finite critical values of a quartic in $K[X]$. In order to make such a question precise, we need to put a height function on the set of such triples; the condition that G_K acts on each of them makes this equivalent to giving a height function on the set of cubics with coefficients in K . Let $H : K^\times \rightarrow [1, \infty)$ be the usual multiplicative Weil height [52, §1.2] and, following Schanuel [41], we work with the relative height $\text{Ht} = H^n$, where $[K : \mathbb{Q}] = n$. In order to uniformize for scaling, the natural choice for our height is the following:

Definition 2.7. Let $\mathcal{S} = \{X_1, X_2, X_3\} \subset \overline{\mathbb{Q}}$ satisfy $(X - X_1)(X - X_2)(X - X_3) = X^3 + aX^2 + bX + c := p(X) \in K[X]$. We set

$$\text{Ht}(\mathcal{S}) = \text{Ht}(p) := \max(\text{Ht}(a)^6, \text{Ht}(b)^3, \text{Ht}(c)^2)$$

and

$$T_K(H) = \{\mathcal{S} = \{X_1, X_2, X_3\} \subset \overline{\mathbb{Q}} \text{ } G_K\text{-stable} : \text{Ht}(\mathcal{S}) \leq H\}.$$

G_K -stable triples and cubics over K with nonzero discriminant biject, so $|T_K(H)| \asymp H^{\frac{2}{6} + \frac{2}{3} + 1} = H^2$, as the vanishing discriminant cuts out a Zariski closed subset of codimension 1. Therefore, the logarithmic density of K -critical triples with respect to our height is

$$(11) \quad \delta_{\text{crit}}(K) := \lim_{H \rightarrow \infty} \frac{\log(\#\{\mathcal{S} = \{X_1, X_2, X_3\} \subset \overline{\mathbb{Q}} : \exists f \in K[X] : C_f = \mathcal{S}, \text{Ht}(\mathcal{S}) \leq H\})}{2 \log H}.$$

We have:

Proposition 2.8. For a number field K , we have

$$\delta_{\text{crit}}(K) \geq \frac{2}{3}.$$

Proof. In virtue of Definition 2.7, we can analogously count (not necessarily short) Weierstrass models (and not isomorphism classes) of elliptic curves over K with height of the model at most H . We focus just on those with j -invariant 0, which is a critical j -invariant over $\mathbb{Q}$. Since the substitution $x \rightarrow x - \frac{a}{3}$ brings the model in short Weierstrass form, any such model has the shape

$$y^2 = p(x) = x^3 + ax^2 + \frac{a^2}{3}x + c, \quad c \neq \frac{a^3}{27}.$$

The condition $\text{Ht}(p) \leq H$ is then equivalent to having both $\text{Ht}(a) \leq H^{\frac{1}{6}}$ and $\text{Ht}(c) \leq H^{\frac{1}{2}}$. This gives $\asymp H^{\frac{1}{3} + 1}$ choices by Schanuel's Theorem, so the numerator in (11) is at least $\frac{4}{3} \log H$. $\square$

Remark 2.8.1. The lower bound of $\frac{2}{3}$ for $\delta_{\text{crit}}(K)$ is likely to be sharp. Indeed, for the similar (but subtly different) problem of counting K -isomorphism classes of elliptic curves over K with a 3-isogeny, the relative density for $K = \mathbb{Q}$ is $\frac{3}{5}$, but it drops to $\frac{5}{18}$ if we remove the $j = 0$ family, see [5, Proposition 4.2]. One expects this family to dominate the count also in our case; however, showing this requires good quantitative control on the amount of cancellation that can happen when changing variables from a minimal model, which seems to be a tricky problem. Moreover, for $K \neq \mathbb{Q}$, the issue of the general lack of a global minimal model makes working with equations even harder.

3. INSIDE THE GALOIS CLOSURE

3.1. Genus 2 covers of critical elliptic curves. The modular curve $Y_0(3)$ is already known to be the reduced Hurwitz space parametrizing classes of maps $\gamma : \mathbb{P}^1 \rightarrow \mathbb{P}^1$ of degree 3 ramifying simply above each of the four branch points (see [22], 4.2). This is the case because these covers are precisely those induced by 3-isogenies of elliptic curves $E_1 \rightarrow E_2$, after quotienting by the respective involutions:

$$\begin{array}{ccc} E & \xrightarrow{\quad} & E' \\ \pi_E \downarrow & & \downarrow \pi_{E'} \\ \mathbb{P}^1 & \xrightarrow{\quad \gamma \quad} & \mathbb{P}^1 \end{array}$$

One may therefore be tempted to make quartic polynomials take up a similar a role, by considering a larger space of maps to a target elliptic curve, and see if they can arise as the induced maps $\mathbb{P}_X^1 \rightarrow \mathbb{P}_x^1$. This is indeed the case:

Definition 3.1. Let $f \in K[X]$ be a good quartic with critical points X_1, X_2, X_3 and let E_f be the associated critical elliptic curve. For $i = 1, 2, 3$, let $f^{-1}(f(X_i)) \setminus \{X_i\} = \{X_{i1}, X_{i2}\}$, and define

$$(12) \quad h_f(X) = \prod_{i=1}^3 (X - X_{i1})(X - X_{i2})$$

to be the *precritical sextic relative to f* .

Remark 3.1.1. Since f is good, h_f has no repeated root. Therefore, the normalization of the curve $C_f : -Y^2 = h_f(X)$ is a smooth genus-2 curve with divisor at infinity supported at two points.

Proposition 3.2. The curve C_f is defined over K . There is a map $\phi_f : C_f \rightarrow E_f$ given by $(X, Y) \mapsto (f(X), \frac{1}{4}Y f'(X))$, for which the diagram

$$\begin{array}{ccc} C_f & \xrightarrow{\phi_f} & E_f \\ \pi_{C_f} \downarrow & & \downarrow \pi_{E_f} \\ \mathbb{P}_X^1 & \xrightarrow{f} & \mathbb{P}_x^1 \end{array}$$

commutes.

Proof. If ϕ_f is well defined then the commutativity follows trivially, so we just need to verify that

$$y^2 = \frac{1}{16}Y^2(f'(X))^2 = -\frac{1}{16}(f'(X))^2 h_f(X) \stackrel{?}{=} -p(f(X)),$$

where p is the polynomial at the right-hand side of (7). Since f has distinct critical values $f(X_i)$, we have

$$p(f(X)) = \prod_{i=1}^3 (f(X) - f(X_i)) = \prod_{i=1}^3 (X - X_i)^2 (X - X_{i1})(X - X_{i2}) = \frac{1}{16}(f'(X))^2 h_f(X).$$

As f and p have coefficients in K , the last equation implies $h_f \in K[X]$, so C_f is defined over K . $\square$

Remark 3.2.1. As π_{C_f} and π_{E_f} have the same degree, C_f is just the fiber product $\mathbb{P}_X^1 \times_{\mathbb{P}_x^1} E_f$, the map $\mathbb{P}_X^1 \rightarrow \mathbb{P}_x^1$ being f . Both points at infinity $\infty^\pm$ are sent to 0_{E_f} by ϕ_f .

Now the question becomes: what does the cover ϕ_f have to do with E_f having a 3-isogeny? As Fried remarks in [22], one-dimensional (reduced) Hurwitz spaces of covers of $\mathbb{P}^1$ are always quotients of the upper half plane $\mathbb{H}$ by finite index subgroups of $\Gamma(1)$ —with the j -line $\Gamma(1) \backslash \mathbb{H}$ taken as parametrizing the branching locus; but they need not be *congruence* modular curves.

In recent work [12], Chen surveys how noncongruence modular curves can be viewed as (components of) moduli spaces $\mathcal{M}(H)$ for H -covers $\phi : C \rightarrow E$ of elliptic curves branched at most above the origin. He remarks that, while the *abelian* case—that of unramified covers, and hence isogenies [12, Proposition 3.4]—clearly yields congruence modular curves, the converse does not hold [12, Proposition 3.4]. The reader can find the precise definition of the stacks $\mathcal{M}(H)$ in [12, §3.2]; they are, in essence, reduced Hurwitz spaces for H -covers of a variable elliptic curve branching only above the origin. Here we use the letter H to denote the Galois group of the cover of the (critical, in our case) elliptic curve, to differentiate from that G of its projection; indeed, both in the dihedral case coming from isogenies and in our own, one has $G \simeq H \rtimes C_2$. In Theorem 3.5, we exhibit an instance of nonabelian Galois group H such that $\mathcal{M}(H)$ is a congruence modular curve, by showing that $\mathcal{M}(A_4) \simeq Y_0(3)$ over $\mathbb{Q}$.

3.2. The Galois closure. As anticipated, we now exploit our construction to directly show why one gets $Y_0(3)$ when parametrizing good quartic polynomials with given critical values. The core idea is to look at the Galois closure X_f of our quartic f :

Proposition 3.3. (i) *There is a map $g_f : X_f \rightarrow C_f$ over K , which makes X_f into the Galois closure of ϕ_f .*

(ii) *There exists an elliptic curve $\tilde{E}_f/K$ and K -rational maps ν_f, ψ_f of degrees 4 and 3 respectively, with ψ_f unramified, such that the following diagram commutes:*

$$\begin{array}{ccccc}
 X_f & \xrightarrow{\nu_f} & \tilde{E}_f & & \\
 & \searrow g_f & & \searrow \psi_f & \\
 & & C_f & \xrightarrow{\phi_f} & E_f \\
 & & \downarrow \pi_{C_f} & & \downarrow \pi_{E_f} \\
 & & \mathbb{P}_X^1 & \xrightarrow{f} & \mathbb{P}_x^1
 \end{array}$$

Proof. (i) We know that $f : \mathbb{P}_X^1 \rightarrow \mathbb{P}_x^1$ has Galois group $G_f \simeq S_4 \simeq A_4 \rtimes C_2$, with the normal subgroup A_4 corresponding to the double cover of $\mathbb{P}_x^1$ obtained by adjoining to the function field $K(x)$ a root of the discriminant. This is simply $y^2 = \text{Disc}_X(f(X) - x)$, which is the critical elliptic curve E_f . Since $C_f \simeq E_f \times_{\mathbb{P}_x^1} \mathbb{P}_X^1$ by Remark 3.2.1, the universal property of fiber products ensures that $X_f \rightarrow \mathbb{P}_x^1$ factors through C_f . We know that $X_f \rightarrow E_f$ is Galois; if it

had a Galois subcover $D \rightarrow E_f$ factoring through C_f , then A_4 would have a normal subgroup contained in C_3 , but it does not.

(ii) Since $A_4 \simeq V_4 \rtimes C_3$ with V_4 the Klein group $C_2 \times C_2$, there is a C_3 -Galois cover

$$\psi_f : X_f/V_4 \rightarrow E_f$$

branching at most over $E_f[2]$. If this cover branched above $P \in E_f(k) \setminus \{0_{E_f}\}$, there would be $Q \in X_f(k)$ with $e_{\pi_{E_f}(P)}(Q) > 2$, contradicting the fact that the inertia is generated by the local monodromy, which is a transposition above $\pi_{E_f}(P)$. Being abelian and branched at most above the origin, ψ_f is an isogeny from another elliptic curve $\tilde{E}_f$ [12, §1.2]. All objects are defined over K since we are inside the Galois closure of f , so we are done. $\square$

We now need the following converse to Propositions 3.2 and 3.3(i):

Lemma 3.4. *Let $\phi : C \rightarrow E$ be a degree 4 cover with Galois group A_4 , where (C, π_C) is a genus 2 **real** hyperelliptic curve and $(E, 0_E, \pi_E)$ is an elliptic curve in Weierstrass form. If $\phi^{-1}(0_E) = \{\infty_C^+, \infty_C^-\}$, the induced map $f_\phi : \mathbb{P}_t^1 \rightarrow \mathbb{P}_x^1$ through π_C and π_E is a good quartic polynomial.*

Proof. The map $f_\phi : \mathbb{P}_t^1 \rightarrow \mathbb{P}_x^1$ totally ramifies above ∞ (and nowhere else, by the Riemann-Hurwitz formula) with only preimage ∞ , so it is a quartic polynomial. It now suffices to show that its Galois closure X_{f_ϕ} factors through E : in that case, by Remark 3.2.1, X_{f_ϕ} factors through the Galois closure X of ϕ ; since this cover has degree 24 over $\mathbb{P}_x^1$ in virtue of our assumption on $\text{Gal}(X/E)$, the two Galois closures must coincide, as $[K(X_{f_\phi}) : K(x)] \leq |S_4| = 24$. Therefore, $K(E) = K(X_{f_\phi})^{A_4}$, which means that E is the discriminant curve of f_ϕ ; as E is smooth, f must then be good by Lemma 2.3. So, let X be the Galois closure of ϕ . As $K(C) = K(E)(t)$, the function field of X is the splitting field of $f_\phi(t) - x$ over $K(E)$, so $K(X) = K(X_{f_\phi})K(E)$. We know that $[K(X) : K(E)] = 12$, so, if $K(E) \not\subset K(X_{f_\phi})$, then $[K(X_{f_\phi}) : K(x)] = \frac{1}{2}[K(X) : K(x)] = 12$. But $K(X_{f_\phi})/K(x)$ is Galois with Galois group $\triangleleft S_4$, so necessarily A_4 . On the other hand, the Galois group contains the inertia group, so in particular the 4-cycle given by local monodromy of f_ϕ at infinity. Since A_4 does not contain a 4-cycle, we are done. $\square$

We can finally show:

Theorem 3.5. *We have $\mathcal{H}_4 \simeq \mathcal{M}(A_4) \simeq Y_0(3)$ over $\mathbb{Q}$.*

Proof. Observe that $\mathcal{H}_4$ is naturally a connected component of $\mathcal{M}(A_4)$ by Proposition 3.3(i). We show that it is the only one: any A_4 -cover $X \rightarrow E_f$ branched only above the origin factors through the C_3 -cover $X/V_4 \rightarrow E_f$, which is a 3-isogeny as remarked in the proof of Proposition 3.3(ii). The Galois cover $X \rightarrow X/V_4$ branches at most over the kernel of the 3-isogeny, with all preimages having the same ramification index e : a priori, either 1, 2 or 4. In the first case, X would have genus 1, making $X \rightarrow E$ an isogeny; this must have abelian monodromy, contradicting the fact that the Galois group is A_4 . The third case is ruled out by the Riemann-Hurwitz formula: we would need to have $2 \mid 3(e-1)$ in the fully branched case. Therefore, X has genus 4 and $X \rightarrow E_f$ has the same ramification as that of X_f : a double transposition above each element of the kernel of the 3-isogeny. In particular, 0_{E_f} has six preimages in X , all with ramification index 2.

Set

$$g : X \rightarrow C = X/C_3, \quad \psi : C \rightarrow E_f.$$

If $g(C) = 1$, then ψ would be unramified, so 0_{E_f} would have 4 preimages in $C(k)$. But then it could not possibly have six in $X(k)$: in that case, at least four of them would have odd ramification index (1 or 3). On the other hand,

$$g(C) \leq \frac{2g(X) - 2}{3} = 2,$$

so C has genus 2 and g is unramified. Therefore, $\psi^{-1}(0_{E_f})$ consists of two points, which are swapped by the hyperelliptic involution. Choosing a model of X —and hence C —so that $\psi^{-1}(0_{E_f}) = \{\infty_C^+, \infty_C^-\}$ (which by definition does not change the class in the reduced Hurwitz space), Lemma 3.4 gives us that $C \rightarrow E_f$ is the lift of a good quartic. We have shown the first isomorphism.

Now, not only A_4 -covers factor through a 3-isogeny, but $\mathcal{M}(A_4) \rightarrow Y(1)$ factors through $\mathcal{M}(C_3) \simeq Y_0(3)$ by the standard theory of orbifold covers [12, p.44]. Since $\mathcal{H}_4 \rightarrow Y(1)$ has degree 4 [35, Proposition 3.4], all curves have degree 4 over the j -line and are therefore isomorphic. $\square$

Having exploited the structure of the Galois closure, it is natural to look at the “completed” diagram—that is, at all the factorizations of $X_f \rightarrow \mathbb{P}_x^1$ through X_f/K , for some $K \leq S_4$. Of course, we can reason up to conjugation, easing the load on terminology.

Recall the interpretation of the Galois closure of a degree d cover as the normalization of any component of its d -fold self-fibered product with the *fat diagonal* Δ removed [23, p. 8.3.2]. In our case, letting

$$(13) \quad C'_f = \mathbb{P}_X^1 \times_{\mathbb{P}_x^1} \mathbb{P}_X^1 \setminus \Delta_f, \quad E'_f = \widetilde{C'_f},$$

E'_f has a degree-3 cover to $\mathbb{P}_X^1$ and we have

$$X_f \simeq (E'_f \times_{\mathbb{P}_X^1} E'_f) \setminus \Delta_{E'_f}.$$

Therefore, $E'_f \simeq X_f / \langle \tau \rangle$, with $\tau \in S_4$ any transposition. The subgroup generated by any double transposition $\langle \{\tau, \tau'\} \rangle$ defines a genus 2 cover

$$Y_f = X_f / \langle \{\tau, \tau'\} \rangle$$

of $\widetilde{E}_f$. Y_f also has a map to X_f/C_4 (since the square of a 4-cycle is a double transposition). Moreover, the extension corresponding to $D_4 \leq S_4$ is that of the cubic resolvent cover $z \mapsto \text{Res}_f(z)$ associated to f .

Remark 3.5.1. The D_4 -extension is also that of the cover in the bottom row of the first diagram of this section for the 3-isogeny ψ_f , so $\tilde{E}_f : y^2 = \text{Res}_f(z)$. So, double covers of cubic resolvents of quartics also parametrize $Y_0(3)$, in the same way that their critical elliptic curves do.

We summarize all this, along with facts we previously worked out, in the following partial table of subgroups of S_4 up to conjugacy, along with their respective quotients of X_f :

Table 1. Subgroups of S_4 up to conjugacy and relative quotients of X_f .

Subgroup	Generators	Quotient	Genus	Jacobian isogeny class
$\{e\}$	e	X_f	4	?
C_2^t (transposition)	$\langle(12)\rangle$	E_f'	?	?
C_2^d (double transposition)	$\langle(12)(34)\rangle$	Y_f	2	?
C_3	$\langle(123)\rangle$	C_f	2	?
C_4	$\langle(1234)\rangle$	?	?	?
$C_2 \times C_2$ (non-normal Klein)	$\langle(12), (34)\rangle$	?	?	?
$V_4 \simeq C_2 \times C_2$ (normal Klein)	$\langle(12)(34), (13)(24)\rangle$	$\tilde{E}_f$	1	E_f
S_3	$\langle(12), (123)\rangle$	$\mathbb{P}_X^1$	0	trivial
D_4	$\langle(1234), (13)\rangle$	$\mathbb{P}^1$	0	trivial
A_4	$\langle(123), (12)(34)\rangle$	E_f	1	E_f
S_4	$\langle(12), (1234)\rangle$	$\mathbb{P}_x^1$	0	trivial

In the next section we will use the theory of *Brauer relations* to complete the table and recover information about the Jacobians of the curves that appear in it.

4. GLOBAL ARITHMETIC OF THE CRITICAL FAMILIES

4.1. Isogenies and Brauer relations. A *Brauer relation* for a finite group G is an equality

$$(14) \quad \sum_i a_i [G/H_i] = \sum_j b_j [G/K_j]$$

of $\mathbb{Z}^+$ -linear combinations of transitive G -sets in the Burnside ring $B(G)$ —that is, such that there is an isomorphism

$$\bigoplus_i \mathbb{Q}[G/H_i]^{a_i} \cong \bigoplus_j \mathbb{Q}[G/K_j]^{b_j}$$

of the corresponding permutation representations as $\mathbb{Q}[G]$ -modules.

By Kani-Rosen theory [13, Theorem 1.5], if X has automorphism group G , then a Brauer relation (14) for G induces an isogeny

$$\bigoplus_i \text{Jac}(X/H_i)^{a_i} \sim \bigoplus_j \text{Jac}(X/K_j)^{b_j}.$$

Let $B(S_4)$ be the vector space of Brauer relations for S_4 and let $\Theta = (\Theta_1, \dots, \Theta_6)$ be its basis described in [19]; setting $R = \Theta M$ for the matrix

$$M = \begin{pmatrix} -2 & 0 & 0 & -4 & 2 & 0 \\ 0 & 2 & 1 & 2 & 0 & -1 \\ 0 & 0 & 1 & 0 & 2 & 1 \\ 0 & 0 & 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & -2 & 1 & 0 \\ -1 & -1 & 0 & -1 & 1 & 0 \end{pmatrix}$$

of determinant 8, we find that the following six elements form a basis for $B(S_4)$:

$$\begin{aligned}
R_1 : [A_4] + 2[D_4] &= 2[S_4] + [V_4] \\
R_2 : 3[A_4] + 2[C_2^t] &= [V_4] + 2[S_3] + 2[A_3] \\
R_3 : [A_4] + [C_2^d] &= [V_4] + [A_3] \\
R_4 : 3[A_4] + 2[S_3] + 2[C_4] &= 4[S_4] + [V_4] + 2[A_3] \\
R_5 : [\{e\}] + 2[S_4] &= [C_2] + [C_3] + [C_4] \\
R_6 : [\{e\}] + 2[C_2^2] &= 2[C_2^t] + [C_2^d]
\end{aligned}$$

R_1 gives $E_f \sim \tilde{E}_f$, which we already knew; R_2 gives

$$(15) \quad E_f^3 \times (\text{Jac}(E'_f))^2 \sim \tilde{E}_f \times \text{Jac}(\mathcal{C}_f)^2 \implies \text{Jac}(\mathcal{C}_f) \sim E_f \times \text{Jac}(E'_f),$$

so, in particular, E'_f has genus 1. R_3 gives

$$E_f \times \text{Jac}(Y_f) \sim \tilde{E}_f \times \text{Jac}(\mathcal{C}_f),$$

so $\text{Jac}(Y_f) \sim \text{Jac}(\mathcal{C}_f) \sim E_f \times \text{Jac}(E'_f)$. R_4 gives

$$E_f^3 \times \text{Jac}(X_f/C_4)^2 \sim \tilde{E}_f \times \text{Jac}(\mathcal{C}_f)^2,$$

so $\text{Jac}(X_f/C_4) \sim \text{Jac}(E'_f)$. R_5 then gives

$$(16) \quad \text{Jac}(X_f) \sim E_f \times \text{Jac}(E'_f)^3$$

and finally R_6 gives $g(X_f/C_2^2) = 0$. This allows us to complete our table 1:

Table 2. Subgroups of S_4 up to conjugacy and relative quotients of X_f .

Subgroup	Generators	Quotient	Genus	Jacobian isogeny class
$\{e\}$	e	X_f	4	$E_f \times \text{Jac}(E'_f)^3$
C_2^t (transposition)	$\langle (12) \rangle$	E'_f	1	$\text{Jac}(E'_f)$
C_2^d (double transposition)	$\langle (12)(34) \rangle$	Y_f	2	$E_f \times \text{Jac}(E'_f)$
C_3	$\langle (123) \rangle$	$\mathcal{C}_f$	2	$E_f \times \text{Jac}(E'_f)$
C_4	$\langle (1234) \rangle$	X_f/C_4	1	$\text{Jac}(E'_f)$
$C_2 \times C_2$ (non-normal Klein)	$\langle (12), (34) \rangle$	$\mathbb{P}^1$	0	trivial
$V_4 \simeq C_2 \times C_2$ (normal Klein)	$\langle (12)(34), (13)(24) \rangle$	$\tilde{E}_f$	1	E_f
S_3	$\langle (12), (123) \rangle$	$\mathbb{P}_X^1$	0	trivial
D_4	$\langle (1234), (13) \rangle$	$\mathbb{P}^1$	0	trivial
A_4	$\langle (123), (12)(34) \rangle$	E_f	1	E_f
S_4	$\langle (12), (1234) \rangle$	$\mathbb{P}_x^1$	0	trivial

In particular, this implies that $\text{Jac}(X_f)$ is isogenous to the product of $\text{Jac}(\mathcal{C}_f)$ with the abelian surface $(E'_f)^2$, and we have

$$(17) \quad L(\text{Jac}(X_f), s) = L(E_f, s)L(E'_f, s)^3 = L(\text{Jac}(\mathcal{C}_f), s)L(E'_f, s)^2.$$

Let us look at the isogeny $\text{Jac}(C_f) \sim E_f \times \text{Jac}(E'_f)$ in more detail. It is customary to refer to a cover $\phi : C \rightarrow E$ of an elliptic curve as *optimal* if, for any factorization $C \rightarrow E_1 \rightarrow E$ of ϕ with E_1 an elliptic curve, $E_1 \rightarrow E$ has degree 1. It is well-known that, in characteristic zero, an optimal genus-2 cover $C \rightarrow E$ of an elliptic curve induces a complementary (optimal) cover $C \rightarrow E'$ of the same degree, call it d , and a splitting $\text{Jac}(C) \rightarrow E \times E'$ which is a (d, d) -isogeny, see [7].

If f is a good quartic, then E_f is an elliptic curve by Lemma 2.3. A_4 has no index 2 subgroup and $\deg \phi_f = 4$, so ϕ_f is optimal; it therefore induces a degree-4 cover from C_f to another elliptic curve D_f over K , such that

$$(18) \quad 0 \rightarrow D_f \rightarrow \text{Jac}(C_f) \rightarrow E_f \rightarrow 0$$

is exact [28, §2, Lemma]. We then have that $E_f \times D_f$ is $(4, 4)$ -isogenous over K to $\text{Jac}(C_f)$, so D_f is isogenous to $\text{Jac}(E'_f)$ by (15). In [25, Lemma 3.1], together with Gallese, Lombardo and Zannier we show how D_f is isomorphic over K to the curve E'_f of (13). In light of this, we will from now on refer to the uniquely determined Prym elliptic curve satisfying (18) as E'_f .

Corollary 4.0.1. *There is an anti-isometry between $E_f[4]$ and $E'_f[4]$, i.e. a G_K -equivariant isomorphism $\iota : E_f[4] \simeq E'_f[4]$ that inverts the Weil pairing.*

In [7], Bruin and Doerksen study genus 2 curves with $(4, 4)$ -split Jacobians. In Section 4 they show how such a curve admits a model $Y^2 = F_1(X)F_2(X)F_3(X)$ with $\{F_1, F_2, F_3\}$ Galois-stable, a so-called quadratic splitting. Notice how in our case such quadratic splitting is given (up to sign) by the factors in (12). If the quadratic splitting is *nonsingular*, they show that the $(4, 4)$ -isogeny $E_1 \times E_2 \rightarrow \text{Jac}(C)$ must factor through a $(2, 2)$ -isogeny to the Jacobian of a genus 2 curve, which in our case can be taken as Y_f in light of the previous table. In [27], Kostantiniou defines *pseudo Brauer relations* and shows that many classes of well-known isogenies are induced by a generalization of Brauer relations. In particular, combining (2) and (5) in [27, Theorem 1.2] with the paragraph after [27, Lemma 4.1] we see how $(4, 4)$ -isogenies of product of elliptic curves to genus-2 Jacobians admitting a nonsingular quadratic splitting are known to be pseudo Brauer verifiable. In our case where C_f is a precritical genus 2 curve and E_f is the relative critical elliptic curve, we have just shown that the $(4, 4)$ -isogeny is also Brauer verifiable. Moreover, we have shown:

Proposition 4.1. *Let $f \in K[X]$ be a good quartic with critical elliptic curve E_f and let $X_f \rightarrow \mathbb{P}_x^1$ be the Galois closure of $f(X) - x$. The isogeny $\text{Jac}(X_f) \sim E_f \times (E'_f)^3$ is Brauer verifiable.*

4.2. The Prym curve. We now work out a Weierstrass model for our Prym elliptic curve E'_f .

Lemma 4.2. *Let $f \in K[X]$ be a good quartic with $f'(X) = 4(X^3 + AX + B)$ and let $j = j(Z_f)$. Then E'_f has j -invariant*

$$(19) \quad j(E'_f) = \frac{j^2}{4(j - 1728)}$$

and the Weierstrass models $y^2 = x^3 - Ax^2 - B^2$ and $y^2 = \prod_{X_i \in Z_f} (x - X_i^2)$.

Proof. The off-diagonal self-fiber product is

$$(20) \quad \frac{f(X) - f(Y)}{X - Y} = X^3 + X^2Y + XY^2 + Y^3 + 2A(X + Y) + 4B = 0.$$

Let $v = X + Y$, $w = X - Y$; then (20) becomes

$$v^3 + vw^2 + 4Av + 8B = 0.$$

Letting $W = vw$, the normalization is

$$(21) \quad W^2 = -v(v^3 + 4Av + 8B),$$

so setting $x = -\frac{2b}{v}$, $y = \frac{bW}{v^2}$ gives the Weierstrass model

$$E'_f : y^2 = x^3 - Ax^2 - B^2.$$

Since $\prod_{X_i \in Z_f} X_i^2 = B^2$, $-\sum_{X_i \in Z_f} X_i^2 = -(\sum X_i)^2 + 2\sum_{i < j} X_i X_j = 2A$ and $\sum_{i < j} X_i^2 X_j^2 = A^2$, the substitution $x \mapsto x + A$ sends it to the other model above. Using the standard formula [15, §1.2] for the j -invariant we get

$$(22) \quad j(E'_f) = \frac{2^{12}A^6}{\Delta(E'_f)} = -\frac{2^{12}A^6}{16B^2(4A^3 + 27B^2)} = -\frac{j^2(4A^3 + 27B^2)}{(432B)^2} = -\frac{j^2}{4(1728 - j)}.$$

□

Remark 4.2.1. In [29, §5.4], Kumar gives a plane model of the Humbert surface of discriminant 16 in terms of the j -invariants of the two elliptic curves to the product of which the parametrized genus-2 Jacobian is $(4, 4)$ -isogenous. Our pair $(j(E_f), j(E'_f))$, as a function of $j = j(Z_f)$, describes a curve in Kumar's (r, s) -plane. Using (19) and [35, Lemma 4.3] one can compute, after substituting

$$t = \frac{j}{256}, \quad s = \frac{2016 - t}{256}, \quad r = -\frac{9(t - 2016)^2}{4096(t - 1728)},$$

that this curve is $9s^2 - 16rs + 18r = 0$.

Say that we wanted to make the 4-congruence ι of Corollary 4.0.1 more explicit. It is easier to see that the two curves admit a 2-congruence, as both are 2-congruent to the critical point curve $\mathcal{E}_f : v^2 = \prod_{X_i \in Z_f} (u - X_i)$. Indeed, we have $E_f[2] = \{\mathcal{O}, (-f(X_i), 0)\}$ by Lemma 2.3 and $E'_f[2] = \{\mathcal{O}, (X_i^2, 0)\}$ by Lemma 4.2. The restriction of ι on 2-torsions is generically compatible with these 2-congruences:

Lemma 4.3. *Let $f \in K[X]$ be a good quartic such that $\text{Gal}(L/K) \simeq S_3$, where L is the splitting field of f' . Then, the only 2 congruence between E'_f and E_f sends X_i^2 to $-f(X_i)$*

Proof. Let $\tilde{\iota}$ be the restriction of our 4-congruence on 2-torsions and let L be the splitting field of f' over K . As an isomorphism of Galois modules, $\tilde{\iota}$ maps identity to identity. On the x -coordinates of points of exact order 2, say that $\tilde{\iota}(X_i^2) = -f(\tau(X_i))$ for some $\tau \in S_3$. By Galois compatibility, for any $\sigma \in G_K$ we have

$$-f(\tau(\sigma X_i^2)) = \tilde{\iota}(\sigma X_i) = -f(\sigma \tau(X_i)),$$

so $\tau \in Z(\text{Gal}(L/K)) = Z(S_3) = \{\text{id}\}$. □

A different way to see that this 2-congruence extends to a 4-congruence is as follows: recall that, under our normalization, $f(X) = (\psi_{3, \mathcal{E}_f}(X) + A^2)/3 = \frac{1}{4}Xf'(X) + AX^2 + 3BX$. Therefore,

$$(23) \quad \begin{aligned} f(X_j) - f(X_i) &= A(X_j^2 - X_i^2) + 3B(X_j - X_i) = (X_j - X_i)(-AX_k + 3B) = \\ &= (X_j - X_i)(X_k^3 - 4X_iX_jX_k) = (X_j - X_i)X_k((X_i + X_j)^2 - 4X_iX_j) = (X_i^2 - X_j^2)(X_i - X_j)^2. \end{aligned}$$

For a curve $C : y^2 = \prod_{i=1}^3 (x - e_i)$, the halves of $T_i = (e_i, 0)$ are the points with $x = e_i + uv$ and $y = \pm uv(u + v)$, where u and v satisfy $u^2 = e_i - e_j$ and $v^2 = e_i - e_k$. Choose square roots δ_{ij} of $X_i^2 - X_j^2$ and η_{ij} of $-f(X_i) + f(X_j)$, and observe that $\eta_{ij} = \pm(X_i - X_j)\delta_{ij}$ by (23). Hence, the map $\pm\delta_{ij} \mapsto \pm(X_i - X_j)\delta_{ij}$ is G_K -equivariant and sends each half of $(X_i^2, 0)$ to an half of $(-f(X_i), 0)$. Notice that there is an ambiguity of sign, so we can have two possible lifts to the 4-torsions. Understanding which one ι is will be relevant in §5. Moreover, the previous argument shows that for any 2-congruence $X_i^2 \mapsto -f(\tau(X_i))$, the condition

$$\frac{f(\tau(X_j)) - f(\tau(X_i))}{X_i^2 - X_j^2} \in (K^\times)^2$$

is equivalent to lifting to a 4-congruence; so, for $\tau \neq \text{id}$, one would need $\frac{X_{\tau(i)}^2 - X_{\tau(j)}^2}{X_i^2 - X_j^2}$ to be a square for all $i \neq j$.

Remark 4.3.1. This strengthens Lemma 4.3, for example extending it to all good quartics over a totally real number field such that $\mathcal{E}_f[2](K) = \{\mathcal{O}\}$. Indeed, under the notation of the lemma, in this case we have $\text{Gal}(L/K) \simeq A_3$, so τ is a 3-cycle, say $i \mapsto i+1 \pmod{3}$ without loss of generality. Thus, we would have a solution $(X_1, X_2, X_3, U, V) \in K^5$ to

$$X_1^2 - X_3^2 = U^2(X_3^2 - X_2^2), \quad X_3^2 - X_2^2 = V^2(X_2^2 - X_1^2)$$

with the first three entries pairwise distinct, implying $U, V \neq 0$ since -1 is not a square. Substituting the second equation in the first then shows that X_1^2 is strictly contained between the other two squares, contradicting both equations.

Observe that, even if our family of derivatives $\{4(X^3 + AX + B)\}$ is a two-parameter one, the (hyper)elliptic fibrations

$$E_f : y^2 = \text{Disc}_X(f(X) - x), \quad C_f : -Y^2 = h_f(X), \quad E'_f : y^2 = x^3 - Ax^2 - B^2,$$

when taken up to K -isomorphism, are one-parameter families over the j -line for $j = j(Z_f)$ (punctured at $j = 0, 1728, \infty$): for $\mathcal{E}_f$, one observes that, generically, good quartics with $j(Z_f) = j$ are linearly equivalent over K , since a $\overline{\mathbb{Q}}$ -automorphism of the critical point curve $\mathcal{E}_f$ preserves the 2-torsion as a Galois module. This fails for $j = 0, 1728$, where one may need to go over a cubic extension because of the extra automorphisms (see the paragraph before [35, Lemma 4.1]). For E_f , linear changes of variable twist it by a fourth power over K in virtue of the properties of the discriminant, thus preserving its K -isomorphism class. They also preserve the K -isomorphism class of C_f : letting $g(X) = f(cX + d) =: f \circ \alpha(X)$, we have

$$f(X) - x_i = (X - X_i)^2(X - X_{i,1})(X - X_{i,2})$$

with x_i the critical values of f , and hence of g . Therefore,

$$\begin{aligned} g(X) - x_i &= f \circ \alpha(X) - x_i = (\alpha(X) - X_i)^2(\alpha(X) - X_{i,1})(\alpha(X) - X_{i,2}) = \\ &= c^4 (X - c^{-1}(X_i - d))^2 (X - c^{-1}(X_{i,1} - d)) (X - c^{-1}(X_{i,2} - d)), \end{aligned}$$

so, by (12), the curve $\mathcal{C}_g$ is isomorphic to $\mathcal{C}_f$ via $(X, Y) \mapsto (cX + d, c^3Y)$. Hence, linear changes of variable must preserve the K -isomorphism class of E'_f as well, in light of (18). Linear rescalings twist the critical elliptic curve by the sixth power of the linear coefficient, thus preserving its K -isomorphism class, and clearly leave the precritical genus 2 curve unchanged.

In virtue of the previous paragraph, let us from now on interchange freely, when $j \neq 0$, between the notation with subscript f to that with subscript j for our families of curves and their maps. Set

$$u := \frac{A^3}{B^2} = \frac{27j}{4(1728 - j)}$$

and notice that $j \mapsto u$ is a bijection. We recall once again that the map $j \rightarrow (E_j, \psi_j)$ gives an isomorphism $\mathbb{A}^1 \simeq Y_0(3)$. The Prym family is not modular, as can be seen by observing how its j -invariant (19) is not any of those in [49, Table 4] (even up to a linear change of variable).

When $j \neq 0$, letting $x \mapsto Ax$ in the model of Lemma 4.2 gives for E'_j the model $y^2 = ux^3 - ux^2 - 1$, and with the additional change of variable $x \mapsto x/u$ we find the K -representative $E'_u : y^2 = x^3 - ux^2 - u^2$ for the Prym family. Consider also the (-1) -twisted family

$$(24) \quad F_j := E'_j{}^{(-1)} : y^2 = x^3 + ux^2 + u^2.$$

Observe that F_j carries the rational section $Q_j = (0, u)$. As $j(F_j)$ has degree 2 over the j -line, it does not factor through any of the maps $X_1(N) \rightarrow X(1)$ for $N > 1$, so the section is generically nontorsion. Observe moreover how $\text{Jac}(C_u^{(-1)})$ has the K -rational section $P_j = [\infty^+ - \infty^-]$.

Proposition 4.4. (i) *We have $Q_j = P_j$ under the embedding $F_j \hookrightarrow \text{Jac}(C_j)^{(-1)}$ obtained by twisting (18).*
(ii) *If $j = 0$, $f(X)$ is linearly equivalent to $g_B(X) = X^4 + 4BX$ for some $B \in K^*/(K^*)^3$. The Prym curve is then*

$$E'_{0,B} : y^2 = x^3 - B^2, \quad B \in K^*/(K^*)^3$$

and the associated twist $F_{0,B}$ has the 3-torsion point $P_{0,B} = (0, B)$.

(iii) *If $j \in \mathbb{Q} \setminus \{0, 1728\}$, Q_j is torsion if and only if $j \in \{-27648/11, 55296/5\}$, in which case it is 5- and 6-torsion, respectively.*

Proof. (i) Recall that we have the map $\phi_j = \phi_f : C_j \rightarrow E_j$ defined in Proposition 3.2, sending both ∞^+ and ∞^- to 0_{E_j} by Remark 3.2.1. Therefore, $P_j \in \text{Jac}(C_j)(K)$ is sent to 0_{E_j} by the push forward ϕ_{j*} ; by (18), this means that P_j lies in $E'_j \subset \text{Jac}(C_j)$. By the paragraph of [25, (13)], P_j has the coordinates $(X, Y) = (\infty, i\infty)$ in the quartic model (21). Following the substitutions after that line, we find the coordinates $P_j = (0, iB)$ on the Weierstrass model of E'_j from Lemma 4.2, hence $P_j = (0, u) \in F_j$.

(ii) The first part of the statement immediately follows from the fact that linear variable changes $X \rightarrow cX$ send g_B to $g_{Bc^{-3}}$ after renormalizing the leading coefficient. The second part follows directly from Lemma 4.2, with the coordinates for $P_{0,B} \in F_{0,B}(K)$ following from the same argument as for (i). It is well-known that $P_{0,B}$ is 3-torsion on $F_{0,B}$.

(iii) Suppose that $Q_j = (0, u)$ is torsion in F_j . Then $nQ_j = 0_{F_j}$ for some $3 \leq n \leq 10$ or $n = 12$, by Mazur's celebrated result [31] (and the fact that Q_j is neither the origin nor 2-torsion, since F_j is in Weierstrass form). The formulas in [45, §3.6] give, for the nine relevant n -division

polynomials $\Psi_n = \Psi_{n,F_j}$ evaluated at Q_j , the following expressions:

$$\begin{aligned}
\Psi_3(Q_j) &= 2^2 u^3 \\
\Psi_4(Q_j) &= -2^5 u^5 \\
\Psi_5(Q_j) &= -2^6 u^8 (u + 4) \\
\Psi_6(Q_j) &= -2^9 u^{12} (u + 8) \\
\Psi_7(Q_j) &= -2^{12} u^{16} (u^2 + 4u - 16) \\
\Psi_8(Q_j) &= 2^{18} u^{21} (u^2 + 8u + 8) \\
\Psi_9(Q_j) &= 2^{20} u^{27} (u^3 + 12u^2 + 64u + 192) \\
\Psi_{10}(Q_j) &= 2^{25} u^{33} (u + 4)(u^3 + 20u^2 + 80u - 64) \\
\Psi_{12}(Q_j) &= -2^{37} u^{48} (u + 8)(3u^4 + 40u^3 + 160u^2 + 256u + 512)
\end{aligned}$$

and hence, after clearing constants and powers of u (which we can do since $j \neq 0 \iff A \neq 0 \iff u \neq 0$),

$$\begin{aligned}
\Psi_3^*(Q_j) &= 1 \\
\Psi_4^*(Q_j) &= 1 \\
\Psi_5^*(Q_j) &= u + 4 \\
\Psi_6^*(Q_j)/\Psi_3^*(Q_j) &= u + 8 \\
\Psi_7^*(Q_j) &= u^2 + 4u - 16 \\
\Psi_8^*(Q_j)/\Psi_4^*(Q_j) &= u^2 + 8u + 8 \\
\Psi_9^*(Q_j)/\Psi_3^*(Q_j) &= u^3 + 12u^2 + 64u + 192 \\
\Psi_{10}^*(Q_j)/\Psi_5^*(Q_j) &= u^3 + 20u^2 + 80u - 64 \\
\Psi_{12}^*(Q_j)/\Psi_6^*(Q_j) &= 3u^4 + 40u^3 + 160u^2 + 256u + 512.
\end{aligned}$$

Since no polynomial of degree larger than 1 appearing at the right-hand side has a rational root, $\Psi_n(Q_j) = 0$ implies either $n = 5 \iff u = -4 \iff j = -27648/11$ or $n = 6 \iff u = -8 \iff j = 55296/5$.

□

Remark 4.4.1. The two twisted Prym curves over $\mathbb{Q}$ (excluding $j = 0$) where the section becomes torsion, $F_{-27648/11}$ and $F_{55296/5}$, are the elliptic curves with Cremona labels 11a3 and 20a2, respectively. The former is $X_1(11)$. They have, among other interesting arithmetic properties, very small conductors. In §5 we explore in more detail how conductors behave in our one-parameter families.

5. LOCAL ARITHMETIC OF CRITICAL ELLIPTIC CURVES

Let $\Delta = \Delta(f') = -(4A^3 + 27B^2)$ and let $\mathcal{E}_f : Y^2 = f'(X)$ be the critical point curve. Our models have respective discriminants

$$(25) \quad \Delta(\mathcal{E}_f) = 16\Delta, \quad \Delta(E'_f) = 16B^2\Delta, \quad \Delta(E_f) = 16B^2\Delta^3$$

by Lemma 4.2 and [35, Proof of Lemma 4.3].

5.1. Bad reduction and Selmer companions. Using the LMFDB [51] to examine some of the curves E_f and E'_f —whose models can be constructed using the scripts in [34]—one observes that, for a prime $\ell > 3$, the two curves appear to have the same reduction type (good, potentially good or potentially multiplicative). The following result confirms it:

Lemma 5.1. *Let $f \in \mathbb{Z}[X]$ be a good quartic and let $\ell \neq 2$ be a prime. E_f and E'_f have the same reduction type at $\ell > 3$.*

Proof. By (25), primes of bad reduction for either curve are contained in $\{2\} \cup \{\ell : \ell \mid \Delta\} \cup \{\ell : \ell \mid B\}$. Recall that, given an integral Weierstrass equation for an elliptic curve, primes ℓ which divide the discriminant with exponent $v_\ell(\Delta) \notin 12\mathbb{Z}$ are necessarily of bad reduction for the curve. One easily checks that bad places $\ell \neq 2, 3$ for E_f are always bad for E'_f : clearly $\ell \mid \Delta(E_f) \implies \ell \mid \Delta(E'_f)$, and if ℓ was good for E'_f then necessarily $\ell \mid A, B$, since rescaling to another integral model makes the discriminant a ℓ -adic unit. But then, $12 \mid v_\ell(\Delta(E'_f))$ would imply $v_\ell(B) \geq 3$ and $v_\ell(A) \geq 2$, so rescaling $x = \ell^2 X$, $y = \ell^3 Y$ (and possibly iterating) gives a model for E'_f minimal at ℓ . From an equation for E_f (say (20) in [35]) it is then immediate that $x = \ell^4 X$, $y = \ell^6 Y$ gives an integral model for E_f minimal at ℓ for which the reduction is good. Viceversa, bad places $\ell \neq 2, 3$ for E'_f are bad for E_f , since again we must have $\ell \mid A, B$, so $v_\ell(B) \geq 3$ and $v_\ell(A) \geq 2$.

Recall that the places ℓ of potential multiplicative reduction for an elliptic curve with j -invariant j are those such that $v_\ell(j) < 0$. Equations (5) and (19) immediately give that the places where E_f and E'_f have potentially multiplicative reduction, with possibly the addition or exception of 2 and 3, are exactly $\{\ell : v_\ell(j - 1728) > 0 \text{ or } v_\ell(j) < 0\}$: this is because $\{\ell : v_\ell(j - 1536) < 0\} = \{\ell : v_\ell(j) < 0\}$ and $\{\ell : v_\ell(j - 1728) > 0\} \cap \{\ell : v_\ell(j) > 0 \text{ or } v_\ell(j - 1536) > 0\} \subset \{2, 3\}$. Since primes of bad reduction are either potentially good or potentially multiplicative, we are done. $\square$

5.2. An infinite family of Selmer companions. Given an elliptic curve E/K , multiplication-by- n gives a short exact sequence of G_K -modules

$$(26) \quad 0 \longrightarrow E[n] \longrightarrow E \xrightarrow{[n]} E \longrightarrow 0.$$

Applying Galois cohomology to (26) yields the exact sequence

$$0 \longrightarrow E(K)/nE(K) \xrightarrow{\kappa_{E,n}} H^1(K, E[n]) \longrightarrow H^1(K, E)[n] \longrightarrow 0,$$

where $\kappa_{E,n}$ is the Kummer map. For every place v of K , the same construction over the completion K_v gives

$$0 \longrightarrow E(K_v)/nE(K_v) \xrightarrow{\kappa_{E,n,v}} H^1(K_v, E[n]) \longrightarrow H^1(K_v, E)[n] \longrightarrow 0.$$

Recall that the n -Selmer group of E/K is

$$(27) \quad \text{Sel}_n(E/K) = \{c \in H^1(K, E[n]) : \text{res}_v(c) \in \text{im}(\kappa_{E,n,v}) \text{ for every place } v \text{ of } K\}.$$

Equivalently,

$$(28) \quad \text{Sel}_n(E/K) = \ker \left(H^1(K, E[n]) \longrightarrow \prod_v H^1(K_v, E) \right),$$

where the v -component of the displayed map is the composite

$$H^1(K, E[n]) \xrightarrow{\text{res}_v} H^1(K_v, E[n]) \longrightarrow H^1(K_v, E).$$

The group $\text{Sel}_n(E/K)$ is finite, see [38] for more details. In [33], Mazur and Rubin introduce the notion of *n-Selmer companions*, any two elliptic curves E_1, E_2 over a number field K with the property that twisting E_1 and E_2 by any given quadratic character of K yields two curves with isomorphic n -Selmer groups. They give some applications, some examples and a useful criterion for determining whether two curves are companions [33, Theorem 3.1].

Until recently, only finitely many pairs of non-isogenous Selmer companions were known (where two pairs related by component-wise simultaneous quadratic twisting are considered the same). In [47], Spencer constructed an infinite family of such pairs by showing that, under certain congruence conditions on the coefficients for a Weierstrass model, a rational elliptic curve and its Hessian are 3-Selmer companions. Here, we show the existence of infinitely many pairs of non-isogenous 2-Selmer companions over $\mathbb{Q}$. We begin with a few useful lemmas (the proof of the first can be found in [42]).

Lemma 5.2. *Let $\ell > 2$ and $u \in 1 + \ell\mathbb{Z}_\ell$. Then u is a 2^k -th power in $\mathbb{Z}_\ell^\times$ for all $k > 0$. Moreover, if $u \in 1 + 2^{k+2}\mathbb{Z}_2$ for some $k > 0$, u is a 2^k -th power in $\mathbb{Z}_2^\times$.*

Lemma 5.3. *Let $f(X) = X^4 + 2AX^2 - 8X$ with $|A| > 7$, and let ℓ be a prime such that $\mathcal{E}_f$, E_f and E'_f are potentially multiplicative over $\mathbb{Q}_\ell$. Then, the 2-congruences between the three curves identify their respective canonical order-2 subgroups (cf. [33, Definition 2.3]). Moreover, ι identifies the canonical order-4 subgroups of E_f and E'_f , assuming additionally that A is odd if $\ell = 2$.*

Proof. [33, Definition 2.3] and [45, VII, Proposition 5.1] imply that the canonical 2-subgroup is generated by the non-nodal Weierstrass point over the extension where the curve is multiplicative, corresponding to $z = -1 \in \overline{\mathbb{Q}_\ell}^\times / q^\mathbb{Z}$ under Tate uniformization. By Remark 4.3.1 the 2-congruences commute, since $\mathcal{E}_f$ has no rational point of exact order two by the rational root theorem, as $|A| \geq 29$. As $B = -2$, (25) implies that coalescence of two Weierstrass roots for either $\mathcal{E}_f$, E_f or E'_f is equivalent to coalescence of the corresponding two roots of the other curves under the respective 2-congruences, so the first claim follows.

For the canonical 4-subgroup, let X_3 be the non-coalescing root of $\mathcal{E}_f$ (and hence X_3^2 and $-f(X_3)$ for E'_f and E_f respectively). The canonical 4-subgroup corresponds to $\mu_4 \subset \overline{\mathbb{Q}_\ell}^\times / q^\mathbb{Z}$ under Tate uniformization, so it is generated by one of the two halves of the non-nodal Weierstrass point, i.e. of $(X_i^2, 0)$ (resp. $(-f(X_i), 0)$) for E'_f (resp. E_f). Recall that for a curve $C : y^2 = \prod_{i=1}^3 (x - e_i)$, the halves of $T_i = (e_i, 0)$ are the points with $x = e_i + uv$, where u and v satisfy $u^2 = e_i - e_j$ and $v^2 = e_i - e_k$. We can distinguish the two halves at the Tate cusp $q = 0$: there, Tate [50, (46)-(48)] gives uniformization $y^2 + xy = x^3$ with smooth locus identified with $\mathbb{G}_m$ via

$$z \mapsto \left(\frac{z}{(1-z)^2}, \frac{z^2}{(1-z)^3} \right).$$

So, the canonical 4-subgroup is generated by the image of $i = \sqrt{-1}$, and we have $x(-1) = -\frac{1}{4}$, $x(i) = -\frac{1}{2}$. For the non-nodal point we then have $uv = -\frac{1}{4}$, but also $u^2 = v^2 = -\frac{1}{4}$ since the node has $x = 0$, so $u = v$. In our case of E'_f and E_f , recalling the notation from the paragraph after Lemma 4.3, this means that ι preserves the canonical 4-subgroups if and only if

$$\frac{X_3 - X_1}{X_3 - X_2} \Big|_{q=0} = 1,$$

and this immediately follows from the coalescence of X_1 and X_2 . $\square$

Lemma 5.4. *Let F be a finite extension of $\mathbb{Q}_\ell$ for some rational prime ℓ , let E_1, E_2 be non-split potentially multiplicative elliptic curves over F with the same quadratic splitting field L/F , and let $\lambda : E_1[2^k] \simeq E_2[2^k]$ identify the canonical order- 2^k subgroups (cf. [33, Definition 2.3]). If the Tate parameters of both curves are non-norms from L , λ identifies the local 2^k -Kummer images.*

Proof. This follows directly from [33, Lemma 5.6], along with Definitions 2.3 and 5.5 in the same paper. $\square$

Theorem 5.5. *Let $A \in 32\mathbb{Z} - 3 \setminus \{-3\}$. For the quartic $f(X) = X^4 + 2AX^2 - 8X$, the elliptic curves E_f and E'_f are 2-Selmer companions.*

Proof. With our choice, we have

$$(29) \quad j(\mathcal{E}_f) = 1728 \frac{A^3}{A^3 + 27}, \quad j(E_f) = -\frac{A^3(A^3 - 216)^3}{(A^3 + 27)^3}, \quad j(E'_f) = -\frac{16A^6}{A^3 + 27}.$$

We want to apply Theorem 3.1 in [33], setting $p = 2$ and $k = 1$. We know from Corollary 4.0.1 that there is a G_K -invariant isomorphism $\iota : E_f[4] \simeq E'_f[4]$, so condition (i) in the theorem is satisfied. Lemma 5.3 shows that (iii) holds, so for (ii) and (iv) to be satisfied it is enough to show that E_f and E'_f have the same set S of primes of potential multiplicative reduction, and $2 \in S$. Observe that $v_3(j(E'_f)) < 0 \iff v_3(j) < 0$ by (29) and $v_3(j(E_f)) < 0 \iff v_3(j) < 0$ or $v_3(j - 1728) > 6$ by (5). Since for us $v_3(j - 1728) = 6 - v_3(A^3 + 27)$ by (29), the two conditions are equivalent, so 3 is a prime of potentially multiplicative reduction for E_f if and only if it is for E'_f . Then, thanks to Lemma 5.1, the requirement reduces to 2 being of potentially multiplicative reduction for both curves. By (29), potential multiplicativity at 2 is equivalent to $v_2(A + 3) > 4$, which follows from our hypothesis on A . $\square$

The infinitude result announced in the Introduction now follows easily:

Proof of Theorem 2. Let us restrict to a maximal subset of the A found in Theorem 5.5 for which the function $A \mapsto (j(E_f), j(E'_f))$ is injective. Clearly such a set is infinite by (29). As E_f and E'_f are non-isogenous over $\mathbb{Q}(j)$ ($j(E'_f)$ has degree 2 and, as we already observed, it is not modular over the j -line), $E_f \sim E'_f$ happens over $\mathbb{Q}$ for finitely many values of j (there are only finitely many possible degrees for cyclic isogenies over $\mathbb{Q}$, by well-known results of Mazur and Kenku [32]), and each such value precludes finitely many A , so the claim follows. $\square$

Lemma 5.6. *Let f be as in the statement of Theorem 5.5 with A coprime to 3. The curves E_f and E'_f have the same conductor.*

Proof. This is equivalent to the two curves having the same bad primes, all of multiplicative reduction. The only primes ℓ dividing their respective discriminants are 2 and those dividing Δ , and they are all of multiplicative reduction: this is tested by checking that ℓ divides the discriminant of a minimal model over $\mathbb{Q}_\ell$ and the quantity c_4 is an ℓ -adic unit [45, VII, Proposition 5.1]. One computes, by using Tate's *Formulaire* [15] (or by recalling $j = \frac{c_4^3}{\Delta}$ and using (5), (19) and (25)):

$$c_4(E_f) = 16A(A^3 - 216), \quad c_4(E'_f) = 16A^2.$$

So, for $\ell > 2$ the models are automatically minimal and, for both curves, if ℓ divides both the discriminant and c_4 , then clearly $\ell \in \{2, 3\}$. Notice that $2^4 \mid c_4$, so the given models may not be minimal over $\mathbb{Q}_2$; still, 2 is of multiplicative reduction for E'_f , as can be seen by computing $c_4 = A^2$ for the model

$$E'_f : Y^2 + XY + Y = X^3 + \frac{5-A}{4}X^2 + \frac{1-A}{4}X - \frac{A+3}{16}$$

obtained through the change of variables $x = 4X + 2$, $y = 8Y + 4X + 4$, which is integral over $\mathbb{Q}_2$ since $16 \mid A + 3$ (and minimal, since $16 \nmid c_4$). The same holds for E_f , since $c_4 = A(A^3 - 216)$ for the model

$$E_f : Y^2 + XY + Y = X^3 + \frac{1-A^2}{2}X^2 + \frac{A^4 - 4A^2 + 72A - 5}{16}X + \frac{A^4 - 8A^3 - 2A^2 + 72A - 447}{64}$$

obtained via $x = 4X + 1$, $y = 8Y + 4X + 4$ —again integral and minimal over $\mathbb{Q}_2$. $\square$

Remark 5.6.1. More generally, $c_4(E_f)$ is an ℓ -adic unit for all bad primes $\ell > 3$ under the weaker hypotheses that $(A, B) = 1$.

Remark 5.6.2. For f as in Lemma 5.6 and with $v_2(A + 3) > 6$, numerical experiments hint to $\mathcal{E}_f$ having conductor N too. We can show this as follows: $\mathcal{E}_f$ has discriminant -16Δ , so bad primes $\ell > 2$ divide $A^3 + 27$, and are all of multiplicative reduction since $c_4(\mathcal{E}_f) = -48A$ and $(A, 3) = 1$. Therefore, we just need to show that $v_2(N(\mathcal{E}_f)) = 1$. We also compute $c_6(\mathcal{E}_f) = 2^6 3^3$ with Tate's *Formulaire* [15], and hence $2^{18} \mid c_4^3 - c_6^2 = -2^{12} 3^3 (A^3 + 27)$. By [48, p. 2.1] (for $p = 2$) and our hypothesis on $v_2(A + 3)$, $\mathcal{E}_f$ is not minimal for $l = 2$, and a minimal model has

$$c_4(\mathcal{E}_f^{\min}) = \frac{c_4}{2^4}, \quad c_6(\mathcal{E}_f^{\min}) = -\frac{c_6}{2^6}, \quad \Delta(\mathcal{E}_f^{\min}) = \frac{-\Delta}{2^{12}}.$$

Using [48, p. 2.1], we find that $v_2(N(\mathcal{E}_f)) = 1 \iff 2 \mid \Delta(\mathcal{E}_f^{\min})$, which is precisely ensured by $v_2(A + 3) > 6$.

In light of Remark 5.6.2, as well as of numerical evidence, it is natural to ask whether the 2-congruences between $\mathcal{E}_f$ and each of E_f, E'_f also induce isomorphisms of 2-Selmer groups. It turns out that this is indeed the case, but under a quadratic twist.

Proposition 5.7. *Let $f(X) = X^4 + 2AX^2 - 8X$, where $A = 2^h d - 3$ with $h \geq 7$ odd and $3 \nmid d \equiv 1 \pmod{4}$. We have*

$$\text{Sel}_2(E_f^{(-1)}/\mathbb{Q}) \simeq \text{Sel}_2(\mathcal{E}_f^{(-1)}/\mathbb{Q}) \simeq \text{Sel}_2(F_f/\mathbb{Q}).$$

Proof. Let us work with $\mathcal{E}_f^{(-1)}$ and F_f : the proof will then extend to $E_f^{(-1)}$ by the equality of local 2-Kummer images provided by Theorem 5.5, which has weaker hypotheses. Looking at the hypotheses of [33, Theorem 6.1], we see that we are lacking conditions (i) and (iii), with (i) being replaced by the 2-congruence $-X_i \mapsto -X_i^2$. Lemma 5.3 shows that the 2-congruence induces an identification of the canonical 2-subgroups.

Now go through the proof of Theorem 6.1: their *Case 2* and *Case 5* are empty for us, and *Case 3* can be followed verbatim since it only requires the 2-congruence. Therefore, we are left with proving that the ℓ -adic 2-Kummer images coincide for $\ell = \infty$ (their *Case 1*) and ℓ of potentially multiplicative reduction (their *Case 4*).

Bad primes: we first prove directly that our curves have the same splitting field $\mathbb{Q}_\ell(\sqrt{-c_6})$ over $\mathbb{Q}_\ell$, replacing Mazur and Rubin's Lemma 5.7(i): equality of splitting fields is equivalent to

$$\gamma_f := \frac{c_6(F_f)}{c_6(\mathcal{E}_f^{(-1)})} \in (\mathbb{Q}_\ell^\times)^2.$$

We have $c_6(F_f) = -2^6(A^3 + 54)$ and $c_6(\mathcal{E}_f^{(-1)}) = -2^6 3^3$, so $\gamma_f = \frac{1}{27}(A^3 + 54)$, which is a square by Hensel lifting since $A^3 + 27 = 0 \pmod{\ell}$ (or, for $\ell = 2$, because $(A^3 + 54)/27 \equiv 1 \pmod{8}$). This automatically settles the case of split multiplicative reduction, where the proof simply uses identification of 2-canonical subgroups on top of equality of splitting field to apply [33, Lemma 5.4].

For the nonsplit case, following Mazur and Rubin's notation we first have to show that either both Tate parameters are norms from the splitting extension, or none is. Let q and q' be the Tate parameters of $\mathcal{E}_f^{(-1)}$ and F_f ; the usual modular identity for the j -invariant gives

$$j = \frac{1}{q} + 744 + 196884q + O(q^2), \quad j(F_f) = \frac{1}{q'} + 744 + 196884q' + O((q')^2),$$

so $q = j^{-1}(1 + \frac{744}{j} + \frac{750420}{j^2} + \dots)$ and the same for q' . The factors in parentheses are in $1 + \ell\mathbb{Z}_\ell$ since $v_\ell(j), v_\ell(j(F_f)) < 0$, and for $\ell = 2$ they are in $1 + 8\mathbb{Z}_2$ in virtue of our hypothesis on $v_2(A + 3)$ and the fact that $4 \mid 744$ and $2 \mid 750420$. Therefore, by Lemma 5.2, they are squares in $\mathbb{Q}_\ell^\times$. Using (29) we get

$$\frac{q}{q'} \in \frac{j(F_f)}{j}(\mathbb{Q}_\ell^\times)^2 = -\frac{A^3}{108}(\mathbb{Q}_\ell^\times)^2 = (\mathbb{Q}_\ell^\times)^2,$$

either by $A^3 + 27 = 0$ for $\ell > 2$ or $A \equiv 5 \pmod{8}$ for $\ell = 2$. Now, set $q/q' = u^2, u \in \mathbb{Q}_\ell$ and let $L = \mathbb{Q}_\ell(\sqrt{3})$ be the common splitting field of $\mathcal{E}_f^{(-1)}$ and F_f . Then, q is a norm from $L^\times$ if and only if q' is, since $u^2 = N_{L/\mathbb{Q}_\ell}(u)$. Therefore, if they are not norms, we are done by Lemma 5.4 for $k = 1$. Moreover, we have $j^{-1} \in 3A(A^3 + 27)(\mathbb{Q}_\ell^\times)^2 \subset -6d(\mathbb{Q}_\ell^\times)^2$ by (29), so the hypothesis $d \equiv 1 \pmod{4}$ is equivalent to j —and hence q —not being a norm from $\mathbb{Q}_2(\sqrt{3})$.

Now suppose that q, q' are norms, so in particular $\ell > 2$. Then, the splitting extension $L/\mathbb{Q}_\ell$ is unramified and hence $v_\ell(q) = -v_\ell(j) \equiv 0 \pmod{2}$, so $v_\ell(A^3 + 27)$ is even. Choose $\beta \in L^\times$ with $N_{L/\mathbb{Q}_\ell}(\beta) = q$ and put $\beta' = \beta/u$, so that $N_{L/\mathbb{Q}_\ell}(\beta') = q'$. The additional generator of the 2-Kummer image provided by [33, Lemma 5.6(ii)] is built by choosing square roots $\alpha^2 = \beta$ and $\alpha'^2 = \beta'$ in $\bar{\mathbb{Q}}_\ell$. So, for the local 2-Kummer images to coincide, it is enough that $u \in (L^\times)^2$: in that case $\alpha/\alpha' \in L$, so the additional generator for $\mathcal{E}_f^{(-1)}$ would differ from that for F_f by the generator of the common subgroup of [33, Lemma 5.6(i)] obtained by uniformizing $\sigma \rightarrow \left(\frac{(\alpha\alpha')^{\sigma\psi(\sigma)}}{\alpha\alpha'}\right)$ for either curve. So, since L is unramified, we just need to check that the valuation of u is even. Notice that $v_\ell(q) = -v_\ell(1/j) = v_\ell(A^3 + 27)$ and, similarly, $v_\ell(q') = v_\ell(A^3 + 27)$; since $v_\ell(u) = \frac{1}{2}v_\ell(q/q')$, we are done.

$\ell = \infty$: we have to show equality of the 2-Kummer images

$$C^{(-1)}(\mathbb{R})/2C^{(-1)}(\mathbb{R}) \rightarrow H^1(G_\mathbb{R}, C^{(-1)})$$

for $C = \mathcal{E}_f, E'_f$, under the identification induced by their 2-congruence. The number of real components of $\mathcal{E}_f^{(-1)}$ is one if f' has one real root and two otherwise. The number of real components of $E_f^{(-1)}$ and F_f is the same as for $\mathcal{E}_f^{(-1)}$ in virtue of the 2-congruence. If it is one we are done,

as multiplication-by-two is surjective on such components, so the 2-Kummer image is trivial. Now assume that f' has three real roots $-X_1 < -X_2 < -X_3$, so we have

$$(30) \quad X_1 + X_2 + X_3 = 0, \quad X_1 X_2 X_3 = 2.$$

Standard 2-descent gives the $(x - T)$ map [39, §5]

$$\phi_f : \mathcal{E}_f^{(-1)}(\mathbb{R})/2\mathcal{E}_f^{(-1)}(\mathbb{R}) \rightarrow (\mathbb{R}^\times/(\mathbb{R}^\times)^2)^3, \quad [(X, Y)] \mapsto ([X + X_1], [X + X_2], [X + X_3]),$$

so the real 2-Kummer image is generated by a point in the bounded component $-X_1 \leq X \leq -X_2$, having signature $\text{sgn}(\phi_f([(X, Y)])) = (+, -, -)$. As the 2-congruence identifies $-X_i$ with $-X_i^2$, we need to show that we get the same signature for the real 2-Kummer image of

$$F_f(\mathbb{R})/2F_f(\mathbb{R}) \rightarrow (\mathbb{R}^\times/(\mathbb{R}^\times)^2)^3,$$

that is, that $X_1^2 > X_2^2, X_3^2$. The two identities in (30) imply $X_2 < 0 < X_1$ and $X_1 = |X_2 + X_3|$, so we are done. □

Remark 5.7.1. The reason why twisting by -1 is useful is in proving equality of real 2-Kummer images, where the argument breaks down for the correspondence $X_i \rightarrow X_i^2$. Moreover, the twist with the marked nontorsion point is precisely F_f , making the statement of Proposition 5.7 useful for exhibiting nontrivial Tate-Shafarevich groups. Still, the same techniques prove the 2-Selmer isomorphism even for the untwisted curves (and the same proof at $\ell = 2$ works even without the congruence condition hypothesis on $d = 2^{-v_2(A+3)}(A+3)$, since the splitting extension is unramified), under the additional hypothesis that $A > -3$, which ensures that there is a single real component. We remark that numerical evidence makes it likely that the hypotheses of odd exponent at 2 and coprimality with 3 can also be removed, but on the other hand the stronger claim of full 2-Selmer companionship with $\mathcal{E}_f$ can be easily disproved experimentally.

5.3. Tate-Shafarevich groups and visibility. Recall that, for any elliptic curve E over a number field K and any positive integer n , we have the exact sequence

$$(31) \quad 0 \rightarrow E(K)/nE(K) \rightarrow \text{Sel}_n(E/K) \rightarrow \text{III}(E/K)[n] \rightarrow 0,$$

where the Tate-Shafarevich group $\text{III}(E/K)$ is defined as

$$\ker \left(H^1(K, E) \xrightarrow{\text{res}_v} \prod_v H^1(K_v, E) \right).$$

Therefore, if two elliptic curves with isomorphic 2-Selmer groups have trivial 2-torsion and different ranks, one of them is forced to have nontrivial Tate-Shafarevich group. Set $r = \text{rk}_{\mathbb{Q}}(F_f)$ and take all ranks over $\mathbb{Q}$.

Corollary 5.7.2. *Let $f(X) = X^4 + 2AX^2 - 8X$ with $A \in \mathbb{Z}$ as in the statement of Theorem 5.5, and let $d \in \mathbb{Q}^\times/(\mathbb{Q}^\times)^2$. We have*

$$(32) \quad \text{rk}(E_f^d) + \dim_{\mathbb{F}_2} \text{III}(E_f^d/\mathbb{Q})[2] = \text{rk}((E'_f)^d) + \dim_{\mathbb{F}_2} \text{III}((E'_f)^d/\mathbb{Q})[2].$$

Moreover, if A is as in the statement of Proposition 5.7, the integer

$$\text{rk}(C^{(-1)}) + \dim_{\mathbb{F}_2} \text{III}(C^{(-1)})[2]$$

is the same for $C = \mathcal{E}_f, E_f, E'_f$. In particular, for $C = E_f$ and A as in the statement of Theorem 5.5, and for $C = \mathcal{E}_f$ and A as in the statement of Proposition 5.7, if $\text{rk}(C^{(-1)}) = 0$ we have

$$(33) \quad \frac{|\text{III}(C^{(-1)}/\mathbb{Q})[2]|}{|\text{III}(F_f/\mathbb{Q})[2]|} \geq 2.$$

Proof. By Theorem 5.5 we have $\text{Sel}_2(E_f^d/\mathbb{Q}) \simeq \text{Sel}_2((E'_f)^d/\mathbb{Q})$ for any $d \in \mathbb{Q}^\times/(\mathbb{Q}^\times)^2$, so (32) follows, provided both curves have rational 2-torsions of the same size; this is indeed the case: we have $E_f^{(-1)}(\mathbb{Q})[2] = F_f(\mathbb{Q})[2] = \{\mathcal{O}\}$ as both 2-torsions are $G_{\mathbb{Q}}$ -isomorphic to $\mathcal{E}_f[2]$, which we know has no finite rational point. The second assertion follows in the same way from Proposition 5.7. In virtue of either of our choices of A , $j(Z_f) = j(\mathcal{E}_f)$ is nonzero and has 2-adic valuation ≤ 1 , so it does not belong to $\{0, -27648/11, 55296/5\}$. Therefore, taking $d = -1$, the positivity of $r = \text{rk}(F_f)$ ensured by Proposition 4.4(iii), together with our hypothesis on $\text{rk}(C^{(-1)})$, implies

$$\dim_{\mathbb{F}_2} \text{III}(C^{(-1)}/\mathbb{Q})[2] = r + \dim_{\mathbb{F}_2} \text{III}(F_f/\mathbb{Q})[2] \geq 1$$

□

Remark 5.7.3. Since F_f has a nontorsion point and, under the hypotheses of Theorem 5.5 (resp. Proposition 5.7) we expect r to have the same parity as $\text{rk}(E_f^{(-1)})$ (resp. $\text{rk}(\mathcal{E}_f^{(-1)})$), it is natural to ask if we can have, say, $\text{rk}(E_f^{(-1)}) > \text{rk}(F_f)$ for f as in Theorem 5.5. This can indeed happen, as standard software can verify for $A = -67363$, in which case $\text{rk}(F_f) = 1$ and $\text{rk}((E_f)^{(-1)}) = 3$.

Remark 5.7.4. The first small values of A as in Theorem 5.5 or which $\text{rk}(E_f^{(-1)}) = 0$ are

$$A = -163, -99, 61, 109, 125, 157, 253.$$

In all cases we can check that $\text{rk}_{\mathbb{Q}}(F_f) = 2$ and $\text{III}(E_f^{(-1)}/\mathbb{Q})[2] \simeq (\mathbb{Z}/2\mathbb{Z})^2$: for $A = -163$, the critical and co-critical curves are not listed on the LMFDB, but we can certify this with the Sage code in [34]; for $A = -99$, E_f has Cremona label 485136h2, while E'_f is 485136b1; for $A = 61$, E_f is 56752e2 and E_f is 56752i1; for $A = 125$, $E_f^{(-1)}$ has Cremona label 244144a1, while F_f is 244144b2; for $A = 157$, $E_f^{(-1)}$ is 101840j2 and $(E'_f)^{(-1)}$ is 101840k1, while for $A = 253$, $E_f^{(-1)}$ is 144592p2 and $(E'_f)^{(-1)}$ is 144592a1.

Another consistent numerical observation is that, for many of the values A considered in Remark 5.7.4, not only $\text{III}(E_f^{(-1)}/\mathbb{Q})[2]$ is nontrivial, but it is also 2-divisible, as its reported sizes on the LMFDB are powers of 4 larger than 4 (for $A = -163$, the same can be certified with the code in [34]). In light of this, one is led to conjecture that, every time the 2-Selmer companionship forces $\text{III}(E_f^{(-1)}/\mathbb{Q})[2]$ to be nontrivial, it also forces its elements to be 2-divisible. The natural explanation compatible with the data is that the following refined version of (32) holds for $d = -1$:

$$(34) \quad 2\text{rk}_{\mathbb{Q}}(E_f^{(-1)}) + \log_2 |\text{III}(E_f^{(-1)}/\mathbb{Q})[4]| = 2\text{rk}_{\mathbb{Q}}(F_f) + \log_2 |\text{III}(F_f/\mathbb{Q})[4]|.$$

In other words, we are led to conjecture that under the hypotheses of Theorem 5.5, our 4-congruence induces an isomorphism of 4-Selmer groups between the (-1) -twists. Moreover, a search using the script in [34] suggests that the weakest condition which picks up this isomorphism is $A \equiv -3 \pmod{32}$ or $A \equiv 45 \pmod{64}$, which we have verified for $|A|$ up to more than 20000:

Conjecture 1. *For $-3 \neq A \equiv -3 \pmod{32}$ or $A \equiv 45 \pmod{64}$ we have*

$$\mathrm{Sel}_4(E_f^{(-1)}/\mathbb{Q}) \simeq \mathrm{Sel}_4(F_f/\mathbb{Q}).$$

Remark 5.7.5. The stronger claim of a 4-Selmer companionship is falsified, even assuming natural hypotheses stronger than those of Proposition 5.7, by taking for example $A = 125$, $d = 443$. Many of the possibilities for the ranks and 4-torsions of Sha allowed by (34) turn out to happen: for instance, for $A = 3037 \equiv -3 \pmod{32}$ we have $\mathrm{rk}_{\mathbb{Q}}(E_f^{(-1)}) = 0$, $\mathrm{rk}_{\mathbb{Q}}(F_f) = 4$ and $\mathrm{III}(E_f^{(-1)}/\mathbb{Q})[4] \simeq (\mathbb{Z}/4\mathbb{Z})^4$.

Remark 5.7.6. For $A = 253 = 2^8 - 3$ it is reported that F_f has rank 2 and $\mathrm{III}(E_f^{(-1)}/\mathbb{Q})$ has size 8^2 , suggesting the existence of elements of exact order 8 in the Tate-Shafarevich group of $E_f^{(-1)}$, and of an isomorphism $\mathrm{Sel}_8(E_f^{(-1)}/\mathbb{Q}) \simeq \mathrm{Sel}_8(F_f/\mathbb{Q})$ (one could turn this suggestion into a formal implication by proving Conjecture 1, or even just by dropping the odd-exponent hypothesis in the next Theorem 5.8). While the latter does not have to be induced by an 8-congruence—let alone one lifting our 4-congruence—it is natural to ask whether there is a trend of $\mathrm{III}(E_f^{(-1)}/\mathbb{Q})$ having elements of exact order an increasingly larger power of 2 as $v_2(A + 3)$ grows. We currently lack sufficient numerical evidence to express a conjecture in this direction. We also remark that the existence of elements of exact order an arbitrarily large power of 2 in the Tate-Shafarevich group of some rational elliptic curve has only been known since the recent breakthroughs of Smith [46] on 2^∞ -Selmer ranks in quadratic twist families.

We are able to prove Conjecture 1 under slightly stronger assumptions on A , which still yield an infinite family:

Theorem 5.8. *Let $A \in 32\mathbb{Z} - 3 \setminus 3\mathbb{Z}$ be such that $v_\ell(A^3 + 27)$ is odd for all $\ell \mid A^3 + 27$ and $(A + 3)2^{-v_2(A+3)} \equiv 1 \pmod{4}$. Then, for $f(X) = X^4 + 2AX^2 - 8X$, we have*

$$\mathrm{Sel}_4(E_f^{(-1)}/\mathbb{Q}) \simeq \mathrm{Sel}_4(F_f/\mathbb{Q}).$$

Proof. We follow the proof structure of Proposition 5.7, still with reference to [33, Theorem 6.1], noting that we are again lacking hypotheses (i) and (iii) in their statement, with (i) being replaced by our twisted 4-congruence (Corollary 4.0.1). The latter maps the canonical 4-subgroups [33, Definition 2.4] to one another for every bad prime in virtue of Lemma 5.3. As in the proof of Proposition 5.7, *Case 2* and *Case 5* are empty for us (the former because we required $(A, 3) = 1$), and *Case 3* is covered by the hypothesis we already have. So, we can focus on bad primes (*Case 4*) and the place at infinity (*Case 1*).

Bad primes: we already know that both curves have potentially multiplicative reduction, multiplicative at $\ell > 2$, and that the splitting extensions $L = \mathbb{Q}_\ell(\sqrt{3})$ agree for both curves. So, equality of 4-Kummer images at split primes follows from [33, Lemma 5.4]. Assume the reduction is not split. Recall the equation

$$q = j^{-1} \left(1 + \frac{744}{j} + \dots \right)$$

for the Tate parameter of an elliptic curve with j -invariant j . The Tate parameter of $E_f^{(-1)}$ (resp. F_f) modulo $(\mathbb{Q}_\ell^\times)^8$ is just $j(E_f)^{-1}$ (resp. $j(E'_f)^{-1}$) by Lemma 5.2. This also holds for $\ell = 2$, because the 2-adic valuation of both j -invariants is at most -3 by (29), in virtue of our hypotheses. By the

same equation, their ratio is

$$\frac{(A^3 - 216)^3}{16A^3(A^3 + 27)^2} \in \frac{(A^3 - 216)^3}{A^3} \mathbb{Q}_\ell^2 = 3^{12} \mathbb{Q}_\ell^2,$$

hence a square; the odd exponent hypothesis implies that both Tate parameters are not norms for $\ell > 2$, since in this case $L/\mathbb{Q}_\ell$ is unramified. Notice that the additional congruence condition on the odd part of $A + 3$ implies that the Tate parameters are not norms at $\ell = 2$ as well, as in the proof of Proposition 5.7. So, we are done by Lemma 5.4 for $k = 2$, since our 4-congruence preserves the canonical 4-subgroups.

$\ell = \infty$: If $\mathcal{E}_f$ has just one real component, then so do $E_f^{(-1)}$ and F_f by the 2-congruences. As multiplication-by-4 is surjective on such a component, both 4-Kummer images are trivial, and we are done. Assume that there are three real components and consider [33, Lemma 5.1]. It tells us that the Kummer image is generated by the standard cocycles attached to the 4-division of the rational 2-torsion points. This cocycle is trivial for the point at infinity and for that in the unbounded real component, as multiplication-by-four is surjective there. For the two points in the bounded component the cocycles differ by a coboundary, so the Kummer image is generated by any of them. Therefore, we just need to show that our 4-congruence sends bounded real component to bounded real component; in other words, we need to show that if $X_1 < X_2 < 0 < X_3$ are the three roots of f' , then $f(X_2) > f(X_1), f(X_3)$. This is because the latter are the Weierstrass roots of $E_f^{(-1)}$ by Lemma 2.3, and for those of F_f we have $-X_2^2 > -X_1^2, -X_3^2$, since $X_3 = -X_1 - X_2$ by our normalization. As $f(X_i) = \frac{1}{4}X_i f'(X_i) + AX_i^2 - 6X_i = AX_i^2 - 6X_i$ and $A = X_3(X_1 + X_2) + X_1X_2 = -X_3^2 + X_1X_2 < 0$, we are done. $\square$

Remark 5.8.1. The set of integers A for which the statement of Theorem 5.8 holds has positive density in $\mathbb{Z}$. This follows by applying the following result of Hooley [26]

Let $g \in \mathbb{Z}[x]$ be a cubic polynomial with the property that for each prime p there exists $n_p \in \mathbb{Z}$ such that $p^2 \nmid g(n_p)$. Then, $g(n)$ is squarefree for a positive proportion of integers n .

to $g(x) = 2^{-v_2(A+3)}(A^3 + 27)$ where, say, $A = 32(3(4x + 1) - 2) - 3$: as $g(0) = 7 \cdot 109$, for all p we can take $n_p = 0$.

In [14], Cremona and Mazur introduce the concept of *visibility* of elements in $H^1(G_K, E) \simeq \text{WC}(E/K)$, showing how it is often the case that nontrivial elements of small order in the Tate-Shafarevich group of rational elliptic curves are *visible* as curves in some abelian surface—usually the Jacobian of a genus 2 curve obtained by gluing E to another elliptic curve F along the graph of a n -congruence $E[n] \simeq F[n]$. These ideas have been developed to show specific (in)visibility results for elements of small order in Sha, for instance in [1], [8] and [21]. Here, we show that the isomorphisms of Selmer groups of Theorems 5.5 and 5.8 lead to visibility of any nontrivial Sha elements *explained* by $F_f(\mathbb{Q})$. In the remainder of this section, let C be either E_f or $\mathcal{E}_f$, and let J be $\text{Jac}(C_f^{(-1)})$ if $C = E_f$ or the abelian surface obtained by glueing F_f with $\mathcal{E}_f^{(-1)}$ along $-X_i \mapsto -X_i^2$, and then quotienting by the common 2-torsion (see [14, Visibility and congruence moduli]). Recall the notation $r := \text{rk}(F_f)$ and set $s = \text{rk}(C^{(-1)})$.

Proposition 5.9. *At least $r - s$ independent generators of $\text{III}(C^{(-1)}/\mathbb{Q})[2^k]/\text{III}(C^{(-1)}/\mathbb{Q})[2^{k-1}]$ are visible in J in all of the following cases:*

- (i) A as in Theorem 5.5, $C = E_f$, $k = 1$;

- (ii) A as in Proposition 5.7, $C = \mathcal{E}_f$, $k = 1$;
- (iii) A as in Theorem 5.8, $C = E_f$, $k = 2$.

Proof. Apply point b) in [14, Visibility and congruence moduli] to $E = C^{(-1)}$, $B = F_f$ and $\iota_{-1} : C^{(-1)}[2^k] \simeq F_f[2^k]$ the $G_{\mathbb{Q}}$ -equivariant isomorphism of $G_{\mathbb{Q}}$ -stable subgroups induced by that of Corollary 4.0.1 (or $X_i \mapsto X_i^2$ if $C = \mathcal{E}_f$). Then, $\sigma \in \text{III}(C^{(-1)}/\mathbb{Q})$ is visible in J if and only if $\iota_{-1}h$ maps to 0 in $H^1(\mathbb{Q}, F_f)$ for any lift $h \in H^1(\mathbb{Q}, C^{(-1)}[2^k])$ of σ —that is, by the standard Kummer exact sequence, if and only if $\iota_{-1}h \in H^1(\mathbb{Q}, F_f[2^k])$ is in the Kummer image of $F_f(\mathbb{Q})/2^k F_f(\mathbb{Q}) \simeq (\mathbb{Z}/2^k\mathbb{Z})^r$. Since $h \in \text{Sel}_{2^k}(C^{(-1)}/\mathbb{Q})$ and the corresponding theorem in the statement shows that ι_{-1} induces an isomorphism $\text{Sel}_{2^k}(C^{(-1)}/\mathbb{Q}) \simeq \text{Sel}_{2^k}(F_f/\mathbb{Q})$, we are done by (31). $\square$

We now give a parity result for the critical elliptic curves of Theorem 5.8.

Lemma 5.10. *For A as in Theorem 5.8 with $128 \mid A + 3$, we have*

$$(35) \quad (-1)^{\text{rk}(E_f^{(-1)})} = -\text{sgn}(A)(-1)^{\omega_1(A^3+27)} \left(\frac{A}{3}\right),$$

where $\omega_1(n)$ counts the number of prime factors of n that are 1 (mod 4).

Proof. By the 2-parity theorem [13, §1], $w(E) = 1$ exactly when $\dim_{\mathbb{F}_2} \text{Sel}_2(E/\mathbb{Q}) - \dim_{\mathbb{F}_2} E(\mathbb{Q})[2]$ is even. Our hypotheses make us fall under those of Proposition 5.7, which implies that the latter quantity is the same for $E = E_f^{(-1)}$, $\mathcal{E}_f^{(-1)}$ (as the 2-torsions are trivial); so, we find

$$w(E_f^{(-1)}) = w(\mathcal{E}_f^{(-1)}).$$

The parity theorem of Dokchitser-Dokchitser for curves with a p -isogeny that are semistable at p [18] implies that $w(E_f^{(-1)}) = (-1)^{\text{rk}(E_f^{(-1)})}$; indeed, any twist of E_f has a 3-isogeny, and we showed in the proof of Theorem 5.8 that the condition $3 \nmid A$ forces good reduction at 3 for $E_f^{(-1)}$.

We use the work of Dokchitser and Cowland-Kellock [13, §2] to compute the root number $w(\mathcal{E}_f^{(-1)})$: it is the product of local root numbers $w_{\ell}(\mathcal{E}_f^{(-1)}) := w(\mathcal{E}_f^{(-1)}/\mathbb{Q}_{\ell})$ for all places ℓ of bad reduction, including ∞ , where it is always -1 . They state that w_2 is $\left(\frac{-c'_6}{4}\right)$ with $c'_6 = c_6/2^{v_2(c_6)}$. In our case, this is $\left(\frac{3^3}{4}\right) = -1$ by Remark 5.6.2. Therefore,

$$w(\mathcal{E}_f^{(-1)}) = \prod_{\ell > 2 \text{ bad}} w_{\ell}(\mathcal{E}_f^{(-1)}).$$

By their Theorem 2.3(iii)-(iv), this is the product of $\left(\frac{c_6}{\ell}\right) = -\left(\frac{3}{\ell}\right)$ over these primes, since they are all multiplicative and the splitting extension is $\mathbb{Q}_{\ell}(\sqrt{-c_6})$. By quadratic reciprocity, this is the same as $(-1)^{\frac{\ell+1}{2}} \left(\frac{\ell}{3}\right)$. Since $\ell \mid A^2 - 3A + 9$ implies $(2A - 3)^2 \equiv -27 \pmod{\ell}$, any such ℓ is 1 (mod 3) again by quadratic reciprocity, so (35) follows from the odd-exponents hypothesis of Theorem 5.8. $\square$

Remark 5.10.1. We could have also deduced Lemma 5.10 from the fact that $c_6(E_f) \in -3\mathbb{Q}_\ell^2$ for $\ell > 3$. Thanks to Remark 5.6.1, this gives the more general formula

$$(36) \quad w(E_f^{(-1)}) = \prod_{\ell|B} -\left(\frac{-1}{\ell}\right) \prod_{\ell|\Delta} -\left(\frac{3}{\ell}\right),$$

under the weaker hypotheses that A is odd, coprime with B and $3 \nmid AB$.

In their work [14], Cremona and Mazur go on to examine specific cases of visibility in abelian surfaces, especially in the case where the n -congruent elliptic curves are semistable of the same conductor N and the visibility occurs inside of $J_0(N)$. For this reason, taking $n = p$ a prime, they define the (necessary) condition of *modular p -congruence*, that is—assuming the curves are *optimal* in their respective isogeny classes—having $E[p] = F[p]$ inside $J_0(N)$. They then remark [14, Proposition] that a sufficient condition for this to happen is (together with a mild congruence condition on $v_p((\Delta_E)$ if $p \mid N$, which in our case translates to $v_2(A+3)$ being odd) that the Fourier coefficients $a_m(E)$, $a_m(F)$ of the cusp forms associated to E and F agree (mod p) for *all* $m > 0$ (as opposed to just those with index coprime to $2N$, which follows just from the 2-congruence).

In our case, under the hypotheses of Lemma 5.6, we are dealing with semistable curves of the same conductor $N = N(A)$. Therefore, it is natural to ask if our 4-congruence, or at least the 2-congruence it induces, can be seen inside $J_0(N)$.

Question 1. In the situation of Lemma 5.6, can the 4-congruence $E_f[4] \simeq E'_f[4]$ be seen inside $J_0(N)$? In other words, does there exist an embedding $\text{Jac}(\mathcal{C}_f) \hookrightarrow J_0(N)$?

Looking at the cusp forms associated to E_f, E'_f for $A = -35, 29$ on the LMFDB, we see that the first twenty coefficients—including those divisible by bad primes—agree mod 2 and even mod 4, so the aforementioned proposition of Cremona and Mazur hints towards a possible affirmative answer. On the other hand, while E'_f is $\Gamma_0(N)$ -optimal in these two cases, E_f is not: therefore, we have to take the optimal representative. Notice that composing the anti-isometry $E'_f[4] \simeq E_f[4]$ with the 3-isogeny $E_f \rightarrow \tilde{E}_f$ yields an anti-isometry $E'_f[4] \simeq \tilde{E}_f[4]$. One may then conjecture that $\tilde{E}_f$ is always optimal, which is the case in all examples we could numerically verify.

5.4. The subfamily $j = 0$. Over the special fiber $A = 0 \iff j(Z_f) = 0$, the critical families E_f, E'_f blow-up to one-parameter families, as already observed in Proposition 4.4(ii) for the Prym family. In this case, letting $f_B(X) = X^4 - 4BX$, the aforementioned proposition shows that $E'_B := (E'_{f_B})^{(-1)} : y^2 = x^3 + B^2$ is the cubic twist by B of $E'_1 : y^2 = x^3 + 1$, while $E_B := E_{f_B}^{(-1)} : y^2 = x^3 + 27B^4$ by [35, Proof of Lemma 4.5]. Here, equality of 2-Selmer groups can still be shown quite easily when B is odd:

Lemma 5.11. *For all odd $B \in \mathbb{Q}^\times/(\mathbb{Q}^\times)^3$, we have $\text{Sel}_2(E_B/\mathbb{Q}) \simeq \text{Sel}_2(E'_B/\mathbb{Q})$.*

Proof. Identify $H^1(\mathbb{Q}_\ell, E_B[2]) \simeq H^1(\mathbb{Q}_\ell, E'_B[2])$ via the (twisted) 4-isometry of Corollary 4.0.1, which we call ι_B . As before, it suffices to show that the local 2-Kummer images $E_B(\mathbb{Q}_\ell)/2E_B(\mathbb{Q}_\ell) \subset H^1(\mathbb{Q}_\ell, E_B[2])$ and $E'_B(\mathbb{Q}_\ell)/2E'_B(\mathbb{Q}_\ell) \subset H^1(\mathbb{Q}_\ell, E'_B[2])$ coincide under this identification. We again follow [33, Proof of Theorem 6.1], keeping in mind that—differently from the situation of Proposition 5.7 and Theorem 5.8—the only bad reduction of our two curves is now additive potentially good (as $j = 0$), at 2, 3 and primes dividing B . So, we can follow verbatim the proof of Mazur and Rubin's

Case 1 and *Case 3*, as well as *Case 2* for $\ell \neq 2$, as none of these depends on the missing hypothesis of multiplicative reduction at $p = 2$.

For $\ell = 2$, we perform 2-descent for E_B and E'_B over $\mathbb{Q}_2$. As every odd (here we use the hypothesis on the parity of B) 2-adic integer is a cube, over $\mathbb{Q}_2$ we have $E'_B \simeq E_1$ and $E_B \simeq E_1^{(3)}$, so it suffices to perform 2-descent for

$$E_1 : y^2 = (x+1)(x+\omega)(x+\omega^2), \quad E_1^{(3)} : y^2 = (x+3)(x+3\omega)(x+3\omega^2),$$

where ω is a primitive cube root of unity. Let $K = \mathbb{Q}_2(\omega)$ and fix an embedding of K in $\mathbb{C}_2$. Since $\mathbb{Q}_2(E_1[2]) = \mathbb{Q}_2(E_1^{(3)}[2]) = K$, we have that, for both $C = E_1, E_1^{(3)}$, the Weil pairing induces an injective map [17, p. 1]

$$w : H^1(\mathbb{Q}_2, C[2]) \rightarrow H^1(\mathbb{Q}_2, \mu_2(\mathbb{Q}_2) \times \mu_2(K)).$$

Composing w , to the left with the Kummer map $\kappa : C(\mathbb{Q}_2)/2C(\mathbb{Q}_2) \rightarrow H^1(\mathbb{Q}_2, C[2])$ and to the right with the Kummer isomorphism given by Hilbert's Theorem 90, gives a map

$$(37) \quad C(\mathbb{Q}_2)/2C(\mathbb{Q}_2) \rightarrow \mathbb{Q}_2^\times/(\mathbb{Q}_2^\times)^2 \times K^\times/(K^\times)^2$$

usually referred to as a $(x-T)$ map [39, §5]. Indeed, it can be described explicitly [17, p. 2]: in our case, it sends the class of $(x, y) \notin C[2]$ to $(x+1, x+\omega)$ for $C = E_1$, and to $(x+3, x+3\omega)$ for $C = E_1^{(3)}$; moreover, it sends the respective $\mathbb{Q}_2$ -rational 2-torsion points $(-1, 0)$ and $(-3, 0)$ to $((-\omega+1)(-\omega^2+1), -1+\omega) = (3, \omega-1)$ and $((-3\omega+3)(-3\omega^2+3), -3+3\omega) = (27, 3\omega-3) \sim (3, 3\omega-3)$, respectively. Recall that—modulo torsion—we have $C(\mathbb{Q}_2) \simeq \mathbb{Z}_2$ (see for example [45, p. VIII]), so $C(\mathbb{Q}_2)/2C(\mathbb{Q}_2) \simeq (\mathbb{Z}/2)^2$; therefore, it is enough to compute (37) for four classes with distinct images, for each of the two curves.

We find four such classes for E_1 and $E_1^{(3)}$ as follows: we already know that the origin maps to $(1, 1)$, as well as where the rational 2-torsion classes map. Taking $x = 4$ and $x = 6$ gives points in $E_1(\mathbb{Q}_2)$ whose respective classes map to $(5, \omega+4)$ and $(7, \omega+6)$, by Lemma 5.2; similarly, taking $x = -7$ and $x = \frac{1}{4}$ gives two points in $E_1^{(3)}(\mathbb{Q}_2)$ whose respective classes map to $(-4, -7+3\omega) \sim (7, -7+3\omega)$ and $(\frac{13}{4}, \frac{1}{4}+3\omega) \sim (5, \frac{1}{4}+3\omega)$. The four Kummer representatives obtained for each curve are pairwise distinct, as $1, 3, 5, 7$ are distinct classes in $\mathbb{Q}_2^\times/(\mathbb{Q}_2^\times)^2$.

It now remains to show that, under the $G_{\mathbb{Q}_2}$ -equivariant isomorphism $\gamma : E_1[2] \simeq E_1^{(3)}[2]$ induced by any of the ι_B , these two sets get mapped to one another. Observe that there are just two possible isomorphisms of $E_1[2]$ and $E_1^{(3)}[2]$ as Galois modules, determined by the image of $(-\omega, 0)$: $(-3\omega, 0)$ or $(-3\omega^2, 0)$. Indeed, the two origins and the two rational 2-torsion points must necessarily map to one another. Since our isomorphism comes from an anti-isometry on the 4-torsions which is compatible with both (twisted) 2-congruences by Lemma 4.3, we can find in which case we are by checking their actions on Weierstrass points: for $\mathcal{E}_f^{(-1)}[2] \simeq F_f[2]$ we have that $-X_i$ is sent to $-X_i^2$, so in our case $-\omega$ is sent to $-\omega^2$; for $\mathcal{E}_f^{(-1)}[2] \simeq E_f^{(-1)}[2]$ we have that $-X_i$ is sent to $f(X_i)$ (recall that there is no $-$ sign in front of f in virtue of Lemma 2.3), so in our case $-\omega$ is sent to $f_1(\omega) = -3\omega$. So, for the resulting congruence $E_1'[2] \simeq E_1[2]$, $-\omega^2$ is sent to -3ω . By construction of the $(x-T)$ map, the automorphism of $\mathbb{Q}_2^\times/(\mathbb{Q}_2^\times)^2 \times K^\times/(K^\times)^2$ induced by (37) is then realized simply by Galois conjugation on the second coordinate. Therefore, it suffices to check that

$$\frac{3\omega-3}{-\omega-2}, \quad \frac{12\omega+1}{3-\omega}, \quad \frac{3\omega-7}{5-\omega}$$

are squares in $K^\times$, which is indeed the case.

□

We refer the reader interested in more details about the root numbers of special twist families of elliptic curves to [16].

6. DIOPHANTINE APPLICATIONS

We now exploit the arithmetical and geometric properties of our families to give some Diophantine applications. We start with some consequences of the local investigation, and move to more diverse ones later in the section.

6.1. Equicritical quartics once again. Looking at the LMFDB pages of various curves E_f for f as in Theorem 5.5, one observes that these curves have not just one rational 3-isogeny, but *two*. In that case, Theorem 3.5 implies that E_f is also a twist of the critical elliptic curve for a quartic g linearly inequivalent to f : we have stumbled once again upon the phenomenon of *equicritical polynomials*. Inequivalent equicritical quartics have been classified in [35], and we indeed see how f as in the statement of Theorem 5.5 is the same quartic as f_t in [35, Theorem 1], for $t = -\frac{A}{3}$. Observe that, because of the transformation law for the discriminant mentioned in the proof of Lemma 2.3, the quartic g_t of [35, Theorem 1]—which is equicritical to $f = f_t$ —has critical elliptic curve $E_f^{(-3)}$.

Take the set of rational quartics, and any of its subsets, as ordered by the maximum modulus of the coefficients. Theorem 5.5 (together with [35, Theorem 1]) shows that, for a positive proportion of normalized quartics $f \in \mathbb{Z}[X]$ admitting an equicritical twin (cf. §2.3), the curves E_f and E'_f are 2-Selmer companions. As our Selmer-isomorphism tools only work inside the equicritical family, it is natural to ask whether there is a direct link between equicriticality and Selmer companionship, but we offer no answer to this. We shall now give an example to illustrate how the critical families behave, for the simple choice of parameter $A = 3$:

Example 6.1. *Let $t = -1$ and $f = f_t$, $g = g_t$ with the notation of [35, Theorem 1]. Then, $E_f = E_g^{(-3)}$ is the elliptic curve with Cremona label 54a1, while E'_f is 864d1 and $(E'_g)^{(-3)}$ is 1728g1. Notice how the latter ones have different j -invariants, -6 and -6^3 respectively, and are both the only curve in their $\mathbb{Q}$ -isogeny class.*

6.2. Szpiro ratios. Recall that the Szpiro ratio of an elliptic curve $E/\mathbb{Q}$ is defined as

$$\sigma(E) = \frac{\log |\Delta_{\min}(E)|}{\log N_E},$$

where $\Delta_{\min}(E)$ is the minimal discriminant of E . The Szpiro conjecture asserts that for any $\epsilon > 0$ there are only finitely many $E/\mathbb{Q}$ with $\sigma(E) > 6 + \epsilon$. There is also a *modified* version [51] of the Szpiro ratio

$$\sigma_m(E) = \frac{\log \max(|c_4|^3, |c_6|^2)}{\log N_E},$$

for which the analog Szpiro conjecture is known to be equivalent to the *abc* conjecture (while the Szpiro conjecture for σ is just implied by *abc*, but not known to imply it).

The semistable curves of Lemma 5.6 appear to exhibit unusually large modified Szpiro ratio, many of which above 6: for instance, the LMFDB reports a modified Szpiro ratio of 6.774044523277239 for $A = 253$. For $v_2(A + 3) > 6$, Remark 5.6.2 tells us that E_f has the same bad primes as $\mathcal{E}_f$, so (25) qualitatively explains why these curves have high σ : while $N = N(E_f) = N(\mathcal{E}_f)$, $\log |\Delta(E_f)| \sim 3 \log(\Delta(\mathcal{E}_f))$ as $A \rightarrow \infty$, so $\sigma(E_f) \sim 3\sigma(\mathcal{E}_f)$.

If we believe in the *abc* conjecture, we would then expect almost all Szpiro ratios of $\mathcal{E}_f$ for f as in Remark 5.6.2 to be $< 2 + \epsilon$ for any $\epsilon > 0$. Actually, notice that $N = \text{rad}(A^3 + 27)$; applying *abc* directly with $a = A^3$, $b = 27$ gives

$$\log(\text{rad}(A)) + \log N = \log(\text{rad}(A(A^3 + 27))) > (1 - \epsilon) \log |\Delta(\mathcal{E}_f)|$$

for almost all A for any $\epsilon > 0$, and hence $\log N > (\frac{2}{3} - \epsilon) \log |\Delta(\mathcal{E}_f)| \implies \sigma(\mathcal{E}_f) < (\frac{3}{2} + \epsilon)$ for almost all A for any $\epsilon > 0$. Therefore, we actually expect almost all Szpiro ratios in our family to be bounded above by $4.5 + \epsilon$ or any $\epsilon > 0$. The reason that many are so high for small A is that choosing $A = 2^k - 3$ forces $\log(\text{rad}(A^3 + 27)) \sim \log(\text{rad}(A^2 - 3A + 9))$, and for small such values of A this can be much lower than $2 \log |A|$, and even than $\log |A|$ (which is what translates to $\sigma(E_f) > 6$).

Another choice of critical point curves $\mathcal{E}_f$ which seems to generate critical elliptic curves E_f with high Szpiro ratios is that of CM elliptic curves. Upon closer examination in the LMFDB, this seems again to happen as a consequence of E_f being semistable at $\ell > 3$. Specifically, for the twist by -3 which has a rational 3-torsion point (Remark 2.4.1), this follows from a theorem of Barrios [4, Theorem 7.1], applied to the family he denotes by C_3 . Indeed, one can easily show that, in our case, his condition for semistability at $\ell \neq 3$ becomes equivalent to $j = j(\mathcal{E}_f)$ being a cube at ℓ , and many CM elliptic curves have j -invariant the cube of an integer. So, for any $j = n^3 \neq 1728$, E_j will have conductor $\text{rad}(n^3 - 1728)$ by (5), up to a bounded power of 3. It is therefore natural to ask how large the minimal discriminant Δ of the critical elliptic curve E_j for such a j can be. One finds $\Delta_{\min}(E_j) = n^3 - 1728$ up to an absolute constant, so

$$\sigma(E_j) = \frac{\log |n^3 - 1728|}{\log \text{rad}(n^3 - 1728)}$$

and once again what we observed for CM critical point curves is a small number phenomenon, since we expect $\text{rad}(n^3 - 1728) \gg_\epsilon n^{2-\epsilon}$ for all $\epsilon > 0$.

6.3. A variables separated equation. Equations of the form

$$f(X) = g(Y)$$

are commonly referred to as *variables separated equations*, and they have a rich literature, see [24]. In the case $f = g$, letting $d = \deg f$, for $d \geq 3$ the equation

$$(38) \quad \frac{f(X) - f(Y)}{X - Y} = 0$$

generically describes a smooth, irreducible affine plane curve, whose normalization has genus $\frac{(d-2)(d-3)}{2}$. Equation (38) was studied by Avanzi and Zannier [3] in the context of solutions in rational functions; over number fields, if one is concerned about the infinitude of solutions, the only nontrivial case is that of quartics, by Faltings's Theorem. Here we settle it generically, in the case $\sqrt{-1} \in K$:

Theorem 6.2. *Let $f \in K[X]$ be a good quartic and let $h_j(X)$ be any associated precritical sextic, where $j = j(Z_f)$.*

- (i) Assume $\sqrt{-1} \in K$. For all but finitely many values of $j \in K$, equation (38) has infinitely many solutions over K for quartics with $j(Z_f) = j$.
- (ii) If $K = \mathbb{Q}$ and $j(Z_f) \neq 0$, equation (38) has infinitely many solutions over $\mathbb{Q}(i)$, unless f is linearly equivalent to $f_{-4}(X) = X^4 - 8X^2 + 16X$ or $f_{-8}(X) = X^4 - 4X^2 + 4X$, in which case there are, respectively, two and three solutions.
- (iii) There are only finitely many specializations $j \in K$ such that the Pell equation

$$(39) \quad R^2 - h_j S^2 = 1$$

has a solution $(R, S) \in \mathbb{C}[X]^2$, but infinitely many such specializations $j \in \mathbb{C}$.

Proof. We notice preliminarily that a solution to (38) over $L \supset K$ defines a point in $E'_f(L)$, and the converse is true with a finite number of exceptions (coming from the normalization). Therefore, there are infinitely many solutions over L if and only if $\text{rk}_L(E'_f) > 0$.

- (i) We have $\text{rk}_K(F_j) > 0$ for all but finitely many j by Proposition 4.4(i) and Silverman's Specialization Theorem [45, Theorem 20.3], and $F_j \simeq_K E'_j$ since $-1 \in (K^\times)^2$, so the claim follows.
- (ii) If $j \neq 0$, again $\text{rk}_{\mathbb{Q}}(F_j) > 0$ unless $u \in \{-8, -4\}$ by Proposition 4.4(iii), and these values corresponds to quartics linearly equivalent to f_{-4} and f_{-8} , given that $u = \frac{A^3}{B^2}$. Since

$$F_{-27648/11}(\mathbb{Q}(i)) \simeq \mathbb{Z}/5 \text{ and } F_{55296/5}(\mathbb{Q}(i)) \simeq \mathbb{Z}/6,$$

as can be checked on the respective LMFDB base-change pages (see also Remark 4.4.1), the finiteness claims follows. As three of the $\mathbb{Q}(i)$ -rational points of E'_j come from the normalization at infinity of the model (38), which is otherwise smooth for good quartics, the solution count also follows.

- (iii) It is known that the Pell equation for h_j (both over $\mathbb{C}(j)[X]$ and over $\mathbb{C}[X]$) is solvable if and only if P_j is torsion (as a section or as a point for fixed j , respectively), see for instance [30, §1.2]. Therefore, by Proposition 4.4(i) and the Silverman specialization Theorem [45, Theorem 20.3], for each number field $L \supset K$ the Pell equation for h_j is solvable over $\mathbb{C}[X]$ only for finitely many $j \in L$.

On the other hand, applying the result [30, Theorem 1.1(b)] of Masser and Zannier with $D = h_j$, $P_D = P_j$, $E_D = F_j$, $n = 1$ gives the second part of the statement, provided we show that F_j , as an elliptic curve over $K(j)$, is not isogenous to a fixed constant curve E_0 . If this was the case, it would follow that, with at most finitely many exceptions, all geometric fibers F_j would be mutually isogenous over $\mathbb{C}$ —the degree of the isogenies being fixed and equal to that of the isogeny $F_j \sim E_0$. Therefore, their j -invariant could assume only finitely many values, contradicting Lemma 4.2.

□

Remark 6.2.1. For $j = 0$ the situation in regards to Theorem 6.2(ii) is mixed: we can get both a finite and an infinite number of solutions to (38), and both for infinitely many such quartics. Indeed, the family $E'_{0,B}$ is that of cubic twists of the Mordell curve $E'_{0,1} : y^2 = x^3 - 1$. It is known [2] that infinitely many have $\mathbb{Q}$ -rank 0 and infinitely many have $\mathbb{Q}$ -rank 1.

6.4. A model for the Galois closure. Our cyclic étale 3-cover $g_f : X_f \rightarrow \mathcal{C}_f$ over K corresponds to the K -rational 3-cyclic subgroup $\ker \hat{\psi}_f$ (cf. Proposition 3.3(ii)) embedded in $\text{Jac}(\mathcal{C}_f)$ via $\phi_f^* : E_f \hookrightarrow \text{Jac}(\mathcal{C}_f)$. Recall from Remark 2.4.1 that the twist $E_f^{(-3)}$ of E_f by -3 has a 3-torsion point:

with this twist applied to the whole diagram, the base-change $g_f^{(-3)} : X_f \rightarrow \mathcal{C}_f^{(-3)}$ of g_f to $K(\sqrt{-3})$ becomes Kummer. As Bruin and Flynn remark in [9, (1)], curves of genus $g = n - 1$ admitting a Kummer n -cover (necessarily of genus g^2 , by the Riemann-Hurwitz formula) over K have a model

$$(40) \quad Y^2 = G(X)^2 + kH(X)^n$$

with $k \in \mathcal{O}_K^\times$, $G, H \in \mathcal{O}_K^\times[X]$ and $\deg G = n$, $\deg H = 2$. The cover is then obtained, up to n -th twist, by taking an n -th root of $Y + G(X)$: in their notation,

$$D_\delta : \delta U^n = Y + G(X), \quad \delta \in K^\times / (K^\times)^3.$$

Observe that, in our (twisted) case, the curve which Bruin and Flynn denote by $\mathcal{F}$ —the quotient of D_δ by any lift of the hyperelliptic involution, when $n - 1$ is even—is just $(E'_f)^{(-3)} = F_f^{(3)}$. In [9, Lemma 2], they show that $\text{Jac}(D_\delta)$ splits as $\text{Jac}(\mathcal{C}) \times \text{Jac}(\mathcal{F})^2$, in accordance with what we already found, and that $D_\delta \rightarrow \mathcal{F}$ has minimal field of definition $K((-k/\delta^2)^{\frac{1}{3}})$; since, in our case, $X_f \rightarrow (E'_f)^{(-3)}$ is defined over $K(\sqrt{-3})$, we obtain that the cubic class of δ is that of k^{-1} .

Given our quartic $f(X) = X^4 + 2AX^2 + 4BX$, a direct computation of the model (40) for $\mathcal{C}_f^{(-3)}$ (which follows, for instance, from the formula $Y^2 = \text{Disc}_{X'} \left(\frac{f(X') - f(X)}{X' - X} \right)$ derived in [25, (14)] for an affine model of $\mathcal{C}_f$) gives:

$$k = -4, \quad G(X) = 4(5X^3 + 9AX + 27B), \quad H(X) = -2(X^2 + 3A).$$

So we find $\delta = 2$, and a model for X_f is given by

$$U^3 G(X) = U^6 + H(X)^3,$$

with the map

$$g_f : X_f \rightarrow \mathcal{C}_f, \quad (X, U) \mapsto (X, -\sqrt{-3}U(U^2 + 2X^3 + 6A))$$

one can now infer from [9, Lemma 1].

6.5. Jacobians isogenous to the power of an elliptic curve. The study of decompositions of genus g Jacobians goes back to Serre and Ekedahl [20], with many open questions still actively researched. In [36], Paulhus tackles the problem of determining for which genera g there is a genus g Jacobian $J = \text{Jac}(C)$ over an algebraically closed field of characteristic 0 with

$$(41) \quad J \sim E^g$$

for an elliptic curve E ; she gives examples for $g \in \{4, 5, 6\}$, with the isogeny being a priori defined over the field of definition of $\text{Aut}(C)$. The only example in the literature of a genus-4 curve whose Jacobian splits as a fourth power over $\mathbb{Q}$ is *Bring's curve*, whose Jacobian is isogenous to $(50a)^4$ as observed by Serre, see [43, §8] [6]. Looking at the splitting (16) of the Jacobian of our Galois closure, it would therefore be of some interest to determine when

$$(42) \quad E_j \sim E'_j,$$

as this would yield other examples of (41) for $g = 4$. Moreover, if we find $j \in K$ such that (42) is defined over K , this would produce an example where (41) is defined over the field of definition of C .

Remark 6.2.2. Denoting by $\mathcal{A}_g$ the moduli space of principally polarized abelian varieties of dimension g and by $\mathcal{T}_g \subset \mathcal{A}_g$ the Torelli locus of Jacobians of smooth curves of genus g , we have

$$\dim \mathcal{A}_g = \frac{g(g+1)}{2}, \quad \dim \mathcal{T}_g = 3g - 3.$$

As the representative E for $\overline{\mathbb{Q}}$ -isomorphism classes of elliptic curves varies, the abelian varieties E^g form a one-dimensional subvariety of $\mathcal{A}_g$. Applying any fixed isogeny correspondence to this family gives another one-dimensional subvariety. Any such translate has expected dimension of intersection $1 + \dim \mathcal{A}_g - \dim \mathcal{T}_g = -\frac{(g-1)(g-4)}{2}$ with $\mathcal{T}_g$. So, $g = 4$ is at the boundary of the unlikely intersection regime, and one expects such a one-dimensional family to meet the Torelli locus in isolated points. Still, there are infinitely many translates, so it is unclear whether we should expect finiteness of such genus-4 Jacobians over $\mathbb{Q}$. See [11] for a discussion of the unlikely intersection case of this problem.

Observe how a first candidate is the subfamily $f(X) = X^4 + 4BX$ with $j = 0$, for which one has $E_f = E_{0,B} : y^2 = x^3 - 27B^4$ and $E'_f = E'_{0,B} : y^2 = x^3 - B^2$, respectively by [35, p.18] and Proposition 4.4(ii). A simple computation shows that these two curves are never isogenous over $\mathbb{Q}$, and the field F of smallest degree where $E_{0,B} \sim E'_{0,B}$ happens for some B is either $\mathbb{Q}(i)$ or $\mathbb{Q}(\sqrt{3})$, over both of which there is such an isogeny for $B \in (F^\times)^3$. Clearly any such B gives the same pair of isogenous curves; in other words, for $f(X) = X^4 + 4X$ we have $\text{Jac}(X_f) \sim E_f^4$ over $\mathbb{Q}(i)$ and $\mathbb{Q}(\sqrt{3})$, but not over $\mathbb{Q}$.

The standard strategy for finding when a genus 2 correspondence yields isogenous elliptic curves over a number field K is to intersect its graph in the affine plane parametrizing the j -invariants of the two curves with that of the modular polynomial Φ_N (for all values of N for which an N -isogeny over K can exist), see [44], [29]. For $K = \mathbb{Q}$, the possible values of N are $N \leq 19$, $N \in \{21, 25, 27, 37, 43, 67, 163\}$. In our case, plugging the j -invariants of E_f and E'_f as functions of j (given by (5) and (19)) in $\Phi_N(X, Y)$ for the required values of N , yields linear factors for $N = 1, 3, 5, 7, 9, 15, 21$. The corresponding j -invariants are listed in the following table (with the exception of $j = 0$, where there is always such an isogeny, but as we have seen it is never defined over $\mathbb{Q}$):

N	j	$(j(E_f), j(E'_f))$
1	2048	$\left(\frac{16384}{5}, \frac{16384}{5}\right)$
3	$\frac{8000}{7}, \frac{55296}{5}$	$\left(\frac{9938375}{21952}, -\frac{15625}{28}\right)$ and $\left(\frac{488095744}{125}, \frac{16384}{5}\right)$
5	-320	$\left(-\frac{121945}{32}, -\frac{25}{2}\right)$
7	$\frac{3375}{2}$	$\left(-\frac{1159088625}{2097152}, -\frac{140625}{8}\right)$
9	-3375	$\left(-\frac{548347731625}{1835008}, -\frac{15625}{28}\right)$
15	270	$\left(\frac{46969655}{32768}, -\frac{25}{2}\right)$
21	-72000	$\left(-\frac{189613868625}{128}, -\frac{140625}{8}\right)$

Observe how, even if we are looking at eight pairs of j -invariants, that of E'_j only takes four values, being shared by two pairs. This is simply a consequence of the fact that E_j has a 3-isogeny, which can be composed with that to E'_j . The table only tells us for which values of the j -invariant j of the critical points we have a K -isogeny between E_f and a *twist* of E'_f : we have to manually check if the twist is the trivial one. This turns out to happen for three of the pairs:

Proposition 6.3. *For $j \in \{\frac{8000}{7}, \frac{3375}{2}, 270\}$ we have*

$$E_j \sim E'_j.$$

over $\mathbb{Q}$.

Proof. Computing models for the curves via the Sage and MAGMA implementations in [34], we find that the three pairs for j as above are, respectively, those with Cremona labels (126a1, 126a3), (162c2, 162c4) and (50a1, 50a4). The LMFDB confirms that first pair is related by a 3-isogeny, the second by a 7-isogeny and the third by a 15-isogeny. $\square$

Our third pair recovers Bring's curve [6], while the other two are new. Using the LMFDB to find models for $\mathcal{E}_f$ and the implementations in [34] for E_f , we obtain Theorem 4.

REFERENCES

- [1] A. Agashe and W.A. Stein. “Visibility of Shafarevich–Tate groups of abelian varieties”. In: *Journal of Number Theory* 97.1 (2002), pp. 171–185. DOI: 10.1006/jnth.2002.2810.
- [2] L. Alpöge, M. Bhargava, and A. Shnidman. *Integers expressible as the sum of two rational cubes*. Preprint, version 3; with an appendix by Ashay Burungale and Christopher Skinner. 2024. arXiv: 2210.10730 [math.NT].
- [3] R.M. Avanzi and U. Zannier. “The equation $f(X) = f(Y)$ in rational functions $X = X(t)$, $Y = Y(t)$ ”. In: *Compositio Mathematica* 139.3 (2003), pp. 263–295. DOI: 10.1023/B:COMP.0000018136.23898.65.
- [4] A.J. Barrios. “Minimal models of rational elliptic curves with non-trivial torsion”. In: *Research in Number Theory* 8, 4 (2022). DOI: 10.1007/s40993-021-00296-4.
- [5] B. Boggess and S. Sankar. “Counting elliptic curves with a rational N -isogeny for small N ”. In: *Journal of Number Theory* 262 (2024), pp. 471–505. DOI: 10.1016/j.jnt.2024.03.004.
- [6] H.W. Braden and L. Disney-Hogg. “Bring’s curve: old and new”. In: *European Journal of Mathematics* 10, 3 (2024). DOI: 10.1007/s40879-023-00706-0. arXiv: 2208.13692 [math.AG].
- [7] N. Bruin and K. Doerksen. “The arithmetic of genus two curves with $(4, 4)$ -split Jacobians”. In: *Canadian Journal of Mathematics* 63.5 (2011), pp. 992–1024. DOI: 10.4153/CJM-2011-039-3. arXiv: 0902.3480 [math.NT].
- [8] N. Bruin and T. Fisher. “Visibility of 4-covers of elliptic curves”. In: *Research in Number Theory* 4, 11 (2018). DOI: 10.1007/s40993-018-0106-1. arXiv: 1701.07528 [math.NT].
- [9] N. Bruin and E.V. Flynn. “ N -covers of hyperelliptic curves”. In: *Mathematical Proceedings of the Cambridge Philosophical Society* 134.3 (2003), pp. 397–405. DOI: 10.1017/S0305004102006448.
- [10] A. Cadoret. “Lifting results for rational points on Hurwitz moduli spaces”. In: *Israel Journal of Mathematics* 164 (2008), pp. 19–59. DOI: 10.1007/s11856-008-0019-0.
- [11] K. Chen, X. Lu, and K. Zuo. *On a question of Ekedahl and Serre*. Preprint. 2017. arXiv: 1703.10377 [math.AG].
- [12] W.Y. Chen. *Noncongruence modular curves as Hurwitz spaces*. Survey preprint. 2025. arXiv: 2510.12003 [math.NT].
- [13] L. Cowland Kellock and V. Dokchitser. “Root numbers and parity phenomena”. In: *Bulletin of the London Mathematical Society* 55.6 (2023), pp. 2557–2597. DOI: 10.1112/blms.12931. arXiv: 2303.07883 [math.NT].
- [14] J.E. Cremona and B. Mazur. “Visualizing elements in the Shafarevich–Tate group”. In: *Experimental Mathematics* 9.1 (2000), pp. 13–28. DOI: 10.1080/10586458.2000.10504633.
- [15] P. Deligne. “Courbes elliptiques: formulaire d’après J. Tate”. In: *Modular Functions of One Variable IV*. Ed. by Bryan J. Birch and Willem Kuyk. Vol. 476. Lecture Notes in Mathematics. Berlin: Springer, 1975, pp. 53–73. DOI: 10.1007/978-3-540-37614-6_4.
- [16] J. Desjardins. “Root number of twists of an elliptic curve”. In: *Journal de Théorie des Nombres de Bordeaux* 32.1 (2020), pp. 73–101. DOI: 10.5802/jtnb.1112. arXiv: 1801.05262 [math.NT].
- [17] Z. Djabri, E.F. Schaefer, and N.P. Smart. “Computing the p -Selmer group of an elliptic curve”. In: *Transactions of the American Mathematical Society* 352.12 (2000), pp. 5583–5597. DOI: 10.1090/S0002-9947-00-02535-6.
- [18] T. Dokchitser and V. Dokchitser. “Parity of ranks for elliptic curves with a cyclic isogeny”. In: *Journal of Number Theory* 128.3 (2008), pp. 662–679. DOI: 10.1016/j.jnt.2007.02.008. arXiv: math/0604149.

- [19] T. Dokchitser and V. Dokchitser. “Regulator constants and the parity conjecture”. In: *Inventiones Mathematicae* 178.1 (2009), pp. 23–71. DOI: 10.1007/s00222-009-0193-7. arXiv: 0709.2852 [math.NT].
- [20] T. Ekedahl and J.P. Serre. “Exemples de courbes algébriques à jacobienne complètement décomposable”. In: *Comptes Rendus de l’Académie des Sciences. Série I. Mathématique* 317.5 (1993), pp. 509–513.
- [21] T. Fisher. *Visible 2-torsion in the Tate–Shafarevich group of an elliptic curve*. Preprint. 2026. arXiv: 2605.31488 [math.NT].
- [22] M.D. Fried. “Arithmetic of 3 and 4 branch point covers: a bridge provided by noncongruence subgroups of $SL_2(\mathbb{Z})$ ”. In: *Séminaire de Théorie des Nombres, Paris 1987–88*. Ed. by Catherine Goldstein. Vol. 81. Progress in Mathematics. Boston: Birkhäuser, 1990, pp. 77–117. DOI: 10.1007/978-1-4612-5788-2_5.
- [23] M.D. Fried. *Riemann’s Existence Theorem: An Elementary Approach to Moduli*. Online book manuscript. In progress. URL: <https://www.math.uci.edu/~mfried/booklist-ret.html>.
- [24] M.D. Fried. “Variables separated equations: strikingly different roles for the Branch Cycle Lemma and the finite simple group classification”. In: *Science China Mathematics* 55.1 (2012), pp. 1–72. DOI: 10.1007/s11425-011-4324-4.
- [25] A. Gallese, D. Lombardo, F. Naccarato, and U. Zannier. *Finding the complement of an elliptic curve inside a Jacobian*. Preprint. 2026. arXiv: 2606.02429 [math.NT].
- [26] C. Hooley. “On the power-free values of polynomials”. In: *Mathematika* 14 (1967), pp. 21–26. DOI: 10.1112/S002557930000797X.
- [27] A. Konstantinou. “Brauer relations, isogenies and parities of ranks”. In: *Journal of Number Theory* 278 (2026), pp. 482–509. DOI: 10.1016/j.jnt.2025.04.016. arXiv: 2311.02137 [math.NT].
- [28] R.M. Kuhn. “Curves of genus 2 with split Jacobian”. In: *Transactions of the American Mathematical Society* 307.1 (1988), pp. 41–49. DOI: 10.1090/S0002-9947-1988-0936803-3.
- [29] A. Kumar. “Hilbert modular surfaces for square discriminants and elliptic subfields of genus 2 function fields”. In: *Research in the Mathematical Sciences* 2, 24 (2015). DOI: 10.1186/s40687-015-0042-9. arXiv: 1412.2849 [math.AG].
- [30] D. Masser and U. Zannier. “Torsion points, Pell’s equation, and integration in elementary terms”. In: *Acta Mathematica* 225 (2020), pp. 227–312. DOI: 10.4310/ACTA.2020.v225.n2.a2.
- [31] B. Mazur. “Modular curves and the Eisenstein ideal”. In: *Publications Mathématiques de l’IHÉS* 47 (1977), pp. 33–186. DOI: 10.1007/BF02684339.
- [32] B. Mazur. “Rational isogenies of prime degree”. In: *Inventiones Mathematicae* 44.2 (1978), pp. 129–162. DOI: 10.1007/BF01390348.
- [33] B. Mazur and K. Rubin. “Selmer companion curves”. In: *Transactions of the American Mathematical Society* 367.1 (2015), pp. 401–421. DOI: 10.1090/S0002-9947-2014-06114-X. arXiv: 1203.0620 [math.NT].
- [34] F. Naccarato. *Code accompanying The Arithmetic of Critical Values*. GitHub repository. URL: <https://github.com/FNacc/Critical-values>.
- [35] F. Naccarato. “The arithmetic of critical values I: equicritical quartic polynomials”. In: *International Mathematics Research Notices* 2026.13, rnag141 (2026). DOI: 10.1093/imrn/rnag141. arXiv: 2501.03244 [math.NT].
- [36] J. Paulhus. “Decomposing Jacobians of curves with extra automorphisms”. In: *Acta Arithmetica* 132.3 (2008), pp. 231–244. DOI: 10.4064/aa132-3-3.
- [37] A. Poirier. “Critical portraits for postcritically finite polynomials”. In: *Fundamenta Mathematicae* 203.2 (2009), pp. 107–163. DOI: 10.4064/fm203-2-2.

- [38] B. Poonen. *The Selmer Group, the Shafarevich–Tate Group, and the Weak Mordell–Weil Theorem*. Expository notes. 2002. URL: <https://math.mit.edu/~poonen/f01/weakmw.pdf>.
- [39] B. Poonen and E.F. Schaefer. “Explicit descent for Jacobians of cyclic covers of the projective line”. In: *Journal für die reine und angewandte Mathematik* 488 (1997), pp. 141–188. DOI: 10.1515/crll.1997.488.141.
- [40] K. Rubin and A. Silverberg. “Families of elliptic curves with constant mod p representations”. In: *Elliptic Curves, Modular Forms, and Fermat’s Last Theorem (Hong Kong, 1993)*. Ed. by John Coates and Shing-Tung Yau. Cambridge, MA: International Press, 1995, pp. 148–161.
- [41] S.H. Schanuel. “Heights in number fields”. In: *Bulletin de la Société Mathématique de France* 107.4 (1979), pp. 433–449. DOI: 10.24033/bsmf.1905.
- [42] J.P. Serre. *Local Fields*. Vol. 67. Graduate Texts in Mathematics. New York: Springer, 1979. DOI: 10.1007/978-1-4757-5673-9.
- [43] J.P. Serre. *Topics in Galois Theory*. Vol. 1. Research Notes in Mathematics. Boston: Jones and Bartlett, 1992.
- [44] T. Shaska. “Curves of genus 2 with (N, N) -decomposable Jacobians”. In: *Journal of Symbolic Computation* 31.5 (2001), pp. 603–617. DOI: 10.1006/jsc.2001.0439.
- [45] J.H. Silverman. *The Arithmetic of Elliptic Curves*. 2nd ed. Vol. 106. Graduate Texts in Mathematics. Dordrecht: Springer, 2009. DOI: 10.1007/978-0-387-09494-6.
- [46] A. Smith. 2^∞ -Selmer groups, 2^∞ -class groups, and Goldfeld’s conjecture. Preprint. 2017. arXiv: 1702.02325 [math.NT].
- [47] H. Spencer. “Constructing families of 3-Selmer companions”. In: *Research in Number Theory* 11, 67 (2025). DOI: 10.1007/s40993-025-00647-5. arXiv: 2502.01351 [math.NT].
- [48] W.A. Stein and M. Watkins. “A database of elliptic curves—first report”. In: *Algorithmic Number Theory, ANTS V*. Ed. by Claus Fieker and David R. Kohel. Vol. 2369. Lecture Notes in Computer Science. Springer, 2002, pp. 267–275. DOI: 10.1007/3-540-45455-1_22.
- [49] A.V. Sutherland and D. Zywna. “Modular curves of prime-power level with infinitely many rational points”. In: *Algebra & Number Theory* 11.5 (2017), pp. 1199–1229. DOI: 10.2140/ant.2017.11.1199.
- [50] J. Tate. “A review of non-Archimedean elliptic functions”. In: *Elliptic Curves, Modular Forms, & Fermat’s Last Theorem (Hong Kong, 1993)*. Ed. by John Coates and Shing-Tung Yau. Vol. 1. Series in Number Theory. Cambridge, MA: International Press, 1995, pp. 162–184.
- [51] The LMFDB Collaboration. *The L-functions and Modular Forms Database*. Online database. URL: <https://www.lmfdb.org>.
- [52] U. Zannier. “Basics on the theory of heights and its applications to certain diophantine problems”. In: *Expositiones Mathematicae* 36.1 (2018), pp. 1–31. DOI: 10.1016/j.exmath.2017.08.008.